GO 3 OF 3 · MODELLING CLIMATE AND ENERGY FUTURES

Research with GCAM

A transport case: from the model's equations to a defensible study, with Saudi Arabia and the Middle East as the running example



INSIDE

- How GCAM 9.1 represents passenger and freight transport, file by file
- A structured review of GCAM and IAM transport studies, incl. GCAM-KSA
- Add-on XML, policies, batch runs, LMDI, scenario discovery, downscaling
- Ten research questions, a project brief and a rubric

Research handbook · checked against gcam-core v9.1 · October 2026

The map of Go 3

Go 1 showed the whole picture; Go 2 opened each component and built a toy version of it. Go 3 is about **research**: changing GCAM on purpose, running it many times, and defending what comes out. The running case is the transport sector, with Saudi Arabia and the Middle East as the regional focus.

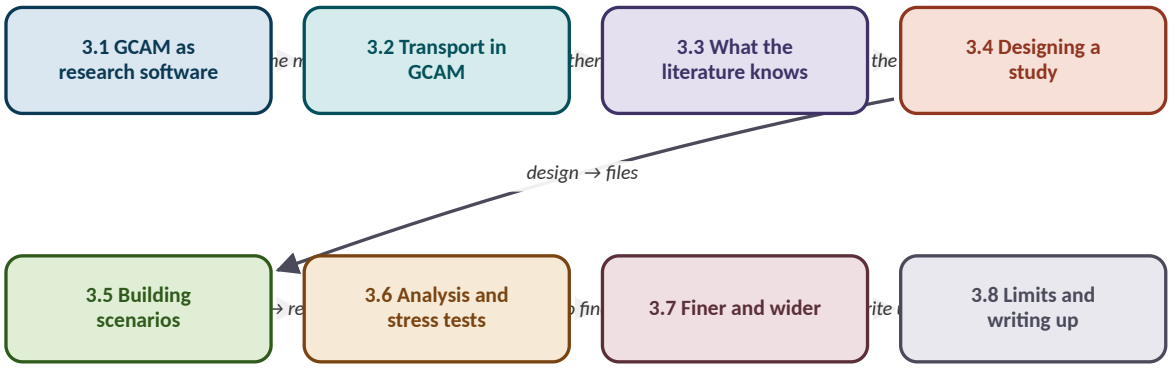


Figure 0 — The eight modules. What to notice: modules 3.1–3.3 are about understanding (the model, the sector, the literature); 3.4–3.6 are the research cycle; 3.7–3.8 take it beyond core GCAM and into print.

BEFORE YOU START

- Go 2 finished, or you can explain logit choice, market clearing and GCAM's components.
- GCAM 9.1 installed and the Reference scenario run and solved (Go 2, module 2.8; lab 7).
- R with rgcam, or Python with gcamreader.
- The Go 3 transport kit folder (`go3-transport-kit/`).

HOW TO USE THIS HANDBOOK

- Read each module with the kit open; every BUILD IT box names a file.
- Do the TRY IT questions before the answers at the back.
- IN GCAM boxes give repository paths (gcam-core v9.1); check them against your own release.
- RESEARCH NOTE boxes mark open questions and pitfalls.

VERSIONS USED

Model: **GCAM 9.1** (tag `gcam-v9.1`, 2 June 2026). Documentation: **v8.2** (JGCRI 2025, doi:10.5281/zenodo.15581183) — older than the code in places, which this handbook points out. Transport units from v8.5: billion pkm/tkm, 1975$, EJ per billion vkm. All paths are relative to the `gcam-core` folder.

What you will be able to do

1. Trace any GCAM transport number to its source file, unit and version, and derive costs, logit shares and demand by hand.
2. Place a GCAM transport result in the literature, with its version and assumptions.
3. Design, build (add-on XML, batch files) and check a scenario study with an uncertainty plan.
4. Decompose, stress-test, downscale or couple results, and write them up so anyone can rebuild every figure.

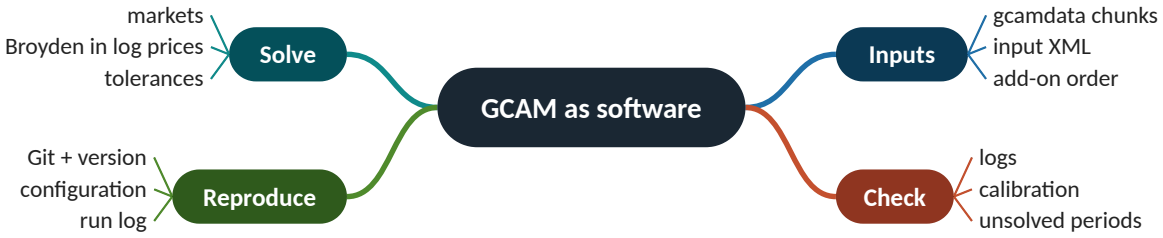
MODULE 1 OF 8

GCAM as research software

Before you change anything, know where every number comes from and how to prove the model solved.

HOOK

GCAM v9.1 was released on 2 June 2026, but its online documentation still describes v8.2. Between the two, the transport input files changed their units from *million* to *billion* passenger-km and from 1990 to 1975 dollars. A number copied from a v8.2 tutorial into a v9.1 add-on file can be wrong by a factor of 1,000, and GCAM will still run. Research with GCAM starts with knowing exactly which version, which file and which unit you are touching.



The pipeline in one picture

GCAM is not one program but a chain of tools. Raw data (IEA energy balances, the UC Davis transport database, FAO land data, SSP population and GDP) is processed by **gcamdata**, an R package, into hundreds of **input XML files**. A **configuration file** lists which XML files to read and in what order. The **C++ core** reads them, solves every market in every period, and writes results to a **BaseX output database**. You read the database with **queries**, from ModelInterface, R (rgcam) or Python (gcamreader).

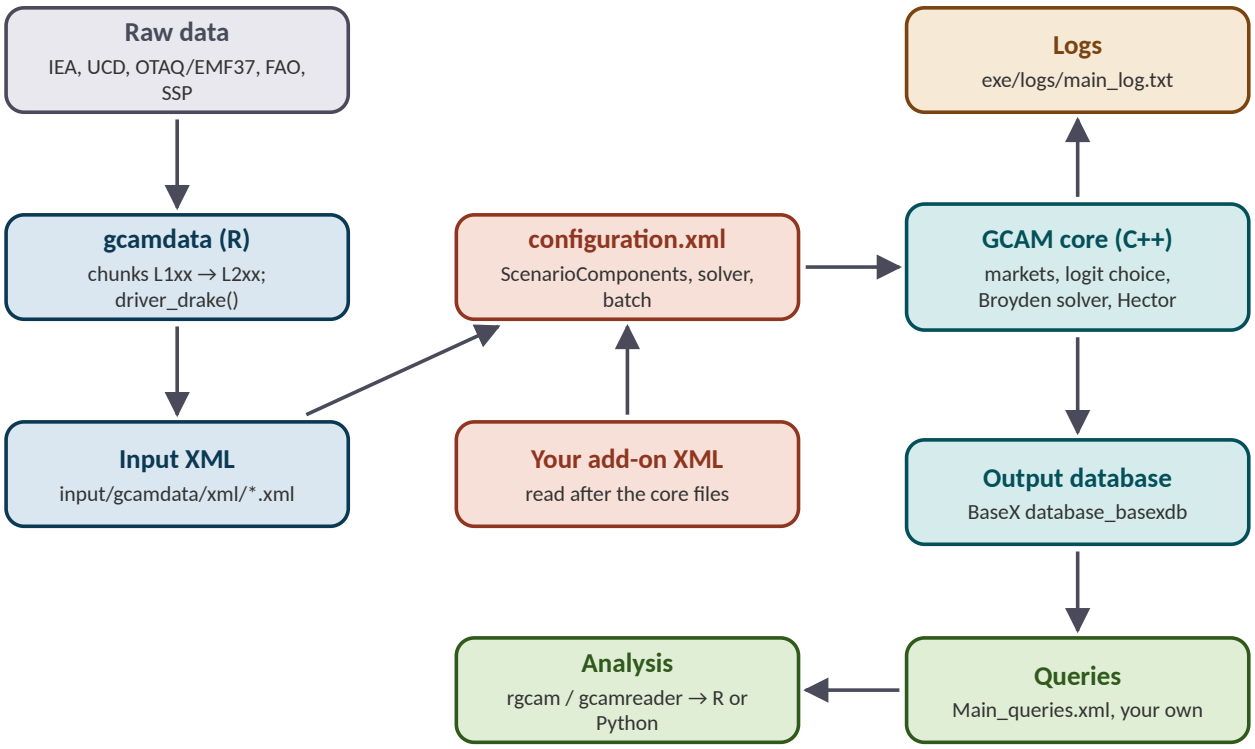


Figure 3.1 — The research architecture. What to notice: your own work enters in only two places, as an add-on XML read after the core files (orange) or as a change inside gcamdata. Everything downstream is the same for every scenario, which is what makes comparisons fair.

Five facts to settle before any study

Item	Current state (checked 23 Oct 2026)	Why it matters
Release	GCAM **9.1**, tag `gcam-v9.1`, 2 June 2026; bundles v8.3–v9.0	Cite the exact version; results change between releases
Documentation	Describes **v8.2** (Zenodo DOI 10.5281/zenodo.15581183)	Some pages lag the code (units, logit types)
Base year	Calibration years 1975, 1990, 2005, 2010, 2015 and **2021**; then 2025–2100 in 5-year steps	The overview page still says 2015; use 2021
Regions	**32** energy-economy regions (Ukraine became region 14 in v8.1); 235 water basins; more than 300 land units	Saudi Arabia sits inside "Middle East" in core GCAM
Transport units	From v8.5: billion pkm/tkm, 1975$, EJ per billion vkm	v8.2 used million pkm, 1990$, BTU per vkm

IN GCAM

Region list: `input/gcamdata/inst/extdata/common/GCAM_region_names.csv` (32 rows in v9.1). Final calibration year: `MODEL_BASE_YEARS` in `input/gcamdata/R/constants.R`. Release notes: `updates.html` in the documentation and the GitHub release page. Saudi Arabia → "Middle East" for transport parameters in `energy/mappings/UCD_ctry.csv`.

How GCAM solves: markets, not optimisation

Every period, GCAM looks for a set of prices at which supply equals demand in every market at once: crude oil, electricity in each region, land in each land unit, water in each basin, and constraint markets such as a CO_2 cap. Write the excess demand of market *m* as a function of all prices **p**:

$$\mathrm{F_m}(\mathbf{p}) = \mathrm{D_m}(\mathbf{p}) - \mathrm{S_m}(\mathbf{p}) = 0 \quad \text{for every market m} \tag{3.1}$$

The solver works in **log prices** and scales each market's excess demand so that a market measured in EJ and one measured in km³ count equally. Its main engine is **Broyden's method**: a Newton-like step that reuses an approximate Jacobian instead of recomputing all derivatives each iteration. Bisection and a preconditioner bring starting guesses close enough for Broyden to converge. A typical regional run solves a few hundred markets.

$$\mathbf{B}_k\,\delta\mathbf{x} = -\mathbf{F}(\mathbf{x}_k), \quad \mathbf{x} = \ln \mathbf{p}, \quad \text{then } \mathbf{B} \text{ is updated from the observed change in } \mathbf{F} \tag{3.2}$$

The model is **recursive-dynamic**: each period is solved with the information of that period only, then the model steps forward five years. Nobody in GCAM knows a carbon price is coming; plants built in 2030 stay until they retire even if a policy arrives in 2035. This is the main contrast with intertemporal optimisation models such as MESSAGEix (Go 2, module 2.8).

SIMPLE CALCULATION

One market, by hand, the way Broyden would

1. Excess demand F(p) = 1000 − 4p − 6p. Guess p_0 = 80: F = 1000 − 800 = 200.
2. Second guess p_1 = 90: F = 1000 − 900 = 100.
3. Secant slope B = (100 − 200) ÷ (90 − 80) = −10.
4. Step δp = −F ÷ B = −100 ÷ −10 = 10, so p_2 = 100.
5. F(100) = 1000 − 1000 = 0: solved. In one dimension Broyden is the secant method; in many dimensions it updates a whole matrix of such slopes.

Answer: price 100, as in the week 4 toy market

Was the run solved? Read the log

A finished GCAM run is not the same as a solved one. Each period's solution is written to `exe/logs/main_log.txt`; a run that completes ends with "Model exiting successfully". If a period fails to reach the tolerance within `max-model-calcs`, the solver reports it and the model continues from an unsolved state. Results from an unsolved period look like any other numbers in the database.

Solver setting (example configuration)	Value	Meaning
`solution-tolerance`	0.001	relative excess demand accepted as "cleared"
`solution-floor`	0.0001	absolute floor for tiny markets
`calibration-tolerance`	0.01	how close base-year outputs must be to data
`max-model-calcs`	2500	budget of model evaluations per period
Broyden `ftol` / `max-iterations`	1e-3 / 25	rule: solution-tolerance ≥ Broyden ftol

Values from the sample solver configuration in the documentation's solver page; the compiled defaults are not given there.

TRAP

"The run finished, so the scenario is valid." Unsolved periods, a constraint that is not binding (and so cannot be solved as a market), or an add-on file silently ignored because of a misspelt region name all produce a database. Every research run needs three checks: the log says each period solved; the base years reproduce calibration data; the variable you changed actually changed in the output (query it directly).

Calibration: what the share-weights are doing

In every base year GCAM must reproduce observed energy use by fuel, sector and region. It does this by solving for **share-weights**: the α_i in the logit share equation (module 3.2). Whatever the costs say, the share-weight adjusts so that the base-year shares match the data. In the future, the share-weights are carried forward or moved along assumed paths (interpolation rules). This is powerful and dangerous: a share-weight silently absorbs everything the cost data miss, including infrastructure, preferences, regulation and data error.

RESEARCH NOTE

Treat future share-weight paths as **assumptions you must report**, not as model output. In the current transport files, battery-electric car share-weights are 0 until the first future year and reach 1 by 2030 along an s-curve; in v5.4 they were 0.25 in 2030 and 0.625 in 2050 (`A54.globaltranTech_shrwt_revised.csv`). The same costs give very different electric-vehicle futures under the two paths. Module 3.6 shows how to test this.

Reproducibility: the four things you must be able to rebuild

What	How	Course file
Model version	Record the release tag and commit hash; never edit the release in place	—
Inputs	Keep every add-on XML and gcamdata change in Git; name scenarios by what they change	lab 12: `gitignore_lab12.txt`
Configuration	One configuration (or batch) file per scenario set, under version control	`lab10_*_config_snippet.xml`, `ensemble_batch.xml`
Analysis	Scripts that read the database and make every figure; no manual edits	labs 12–13 helper and solution scripts

BUILD IT

A minimal run log, kept next to the results (one line per run):

```
date,       gcam_tag,  commit,  config,                    scenario,        solved,  notes
2026-11-02, gcam-v9.1, 11e128f, configuration_trn_ref.xml, Ref_SSP2,        yes,
baseline
2026-11-02, gcam-v9.1, 11e128f, configuration_trn_bev.xml, BEVcost_lo,      yes,     -30%
BEV non-energy cost from 2030
2026-11-03, gcam-v9.1, 11e128f, batch_transport.xml,       Ens_*,           35/36,
Ens_solHi_ccsLo failed 2085
```

TRY IT NOW

1. Open `exe/configuration_ref.xml` in your release and find the line that loads the transport file. Which file is it, and what comes immediately after it? (Answer key.)
2. Explain in two sentences why an add-on file must be listed *after* `transportation_UCD_CORE.xml`.

LOOK IT UP

- GCAM documentation (v8.2): jgcri.github.io/gcam-doc/toc.html — overview, solver, user guide.
- GCAM releases: github.com/JGCRI/gcam-core/releases (v9.1, 2 June 2026).
- Calvin et al. (2019) GCAM v5.1, *Geosci. Model Dev.* 12, 677–698, doi:10.5194/gmd-12-677-2019.
- Bond-Lamberty et al. (2019) gcamdata, *J. Open Research Software*, doi:10.5334/jors.232.

RECAP SO FAR

1. **3.1 GCAM as research software:** data → gcamdata → XML → C++ core → database → queries; add-ons load in order; check the log before you trust a number.

MODULE 2 OF 8

Transport in GCAM, in full

Four demands, five nesting levels, one generalised cost: how GCAM decides how people and goods move.

HOOK

For a Saudi driver in our worked example, owning and running a compact petrol car costs about **$0.29** per passenger-km, of which fuel is only **$0.014**. The **time** spent driving, valued at a wage, costs almost as much as the car itself: **$0.23** per passenger-km. In GCAM, cheap fuel barely moves the mode choice; speed and income do.



What is in the box

Transport in GCAM has four **final demands**, each a service rather than an energy quantity. Energy enters only through the vehicles that provide the service.

Final demand	Service unit (v8.5 on)	v8.2 unit	Per-capita based	Price elasticity	Income elasticity (core / SSP1)
`trn_pass`	billion pass-km	million pass-km	yes	−1.25	1.0 / 0.8
`trn_freight`	billion ton-km	million ton-km	no	−0.65	0.75 / 0.5
`trn_aviation_intl`	billion pass-km	million pass-km	yes	−1.0	1.0 / 1.0
`trn_shipping_intl`	billion ton-km	million ton-km	no	−0.65	0.4 / 0.4

Source: `input/gcamdata/inst/extdata/energy/A54.demand.csv` and `A54.demand_ssp1.csv` (v9.1); unchanged since v5.4. Pipelines and off-road machinery are outside transport; since v6.0 there is a separate `Off_road` industrial sector.

The nesting tree

Each demand is met by a tree of **supplysectors**, **tranSubsectors** (modes or size classes) and **tranTechnologies** (drivetrains). At each branch point a logit shares the market among the branches by cost. Pass-through nodes ("road", "LDV", "4W") simply hand the choice down one level.

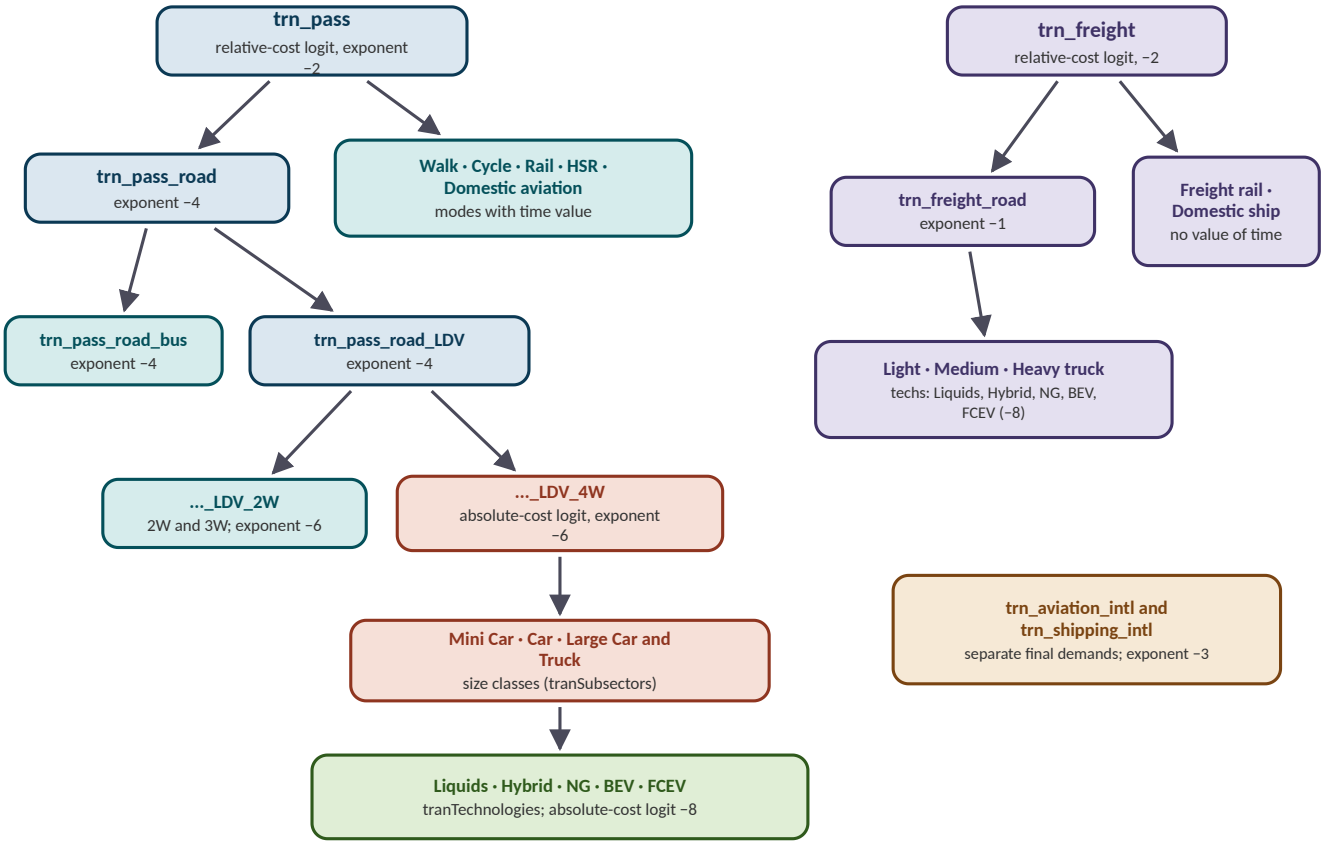


Figure 3.2 — Transport nesting in GCAM v9.1 (from `A54.sector.csv`, `A54.tranSubsector_logit_revised.csv` and `A54.globaltech_passthru_revised.csv`). What to notice: the exponents get more negative further down — buyers switch easily between a petrol and a battery Car (−8) but reluctantly between road and rail (−2).

Mode / class	Technologies (v9.1)	Fuel input markets
Cars (Mini Car, Car, Large Car and Truck)	Liquids, Hybrid Liquids, NG, BEV, FCEV	refined liquids enduse, delivered gas, elect_td_trn, **H2 LDV**
2W and 3W	Liquids, NG, BEV (LA-BEV share-weight 0)	as above
Bus; Light/Medium/Heavy truck	Liquids, Hybrid Liquids, NG, BEV, FCEV	FCEV uses **H2 MHDV**
Passenger rail · HSR	Liquids, Electric · Electric	elect_td_trn
Freight rail	Coal (phased out), Liquids, Electric, BEV, FCEV, Hybrid	FCEV uses H2 MHDV
Domestic and international aviation	Liquids, BEV, Hydrogen	Hydrogen uses **LH2**
Domestic and international ship	Liquids, Hybrid Liquids, BEV, FCEV	FCEV uses LH2
Walk, Cycle	non-motorised	none (speed 4.8 and 15 km/h)

TRAP

Biofuels, e-fuels and ammonia are not transport technologies. Low-carbon liquids reach vehicles only through the `refining` sector, which blends oil, gas- and coal-to-liquids, ethanols, biodiesel and FT biofuels into one pool, `refined liquids enduse`. Aviation and shipping draw on the same pool as cars, so a sustainable-aviation-fuel mandate needs a model change. Core v9.1 has no power-to-liquids or DAC-to-liquids technology, and ammonia is an industrial product, not a transport fuel.

Step 1: the cost of a technology per passenger-km

Start with one vehicle. It burns *I* units of fuel per vehicle-km at fuel price P_f, costs *N* per vehicle-km to own and maintain, and carries *L* people on average (the **load factor**). The cost of one passenger-km is the cost of one vehicle-km shared among the passengers:

$$P_j = (P_f \cdot I_j + N_j) \div L_j \tag{3.3}$$

In the code this is `TranTechnology` (`cvs/objects/technologies/source/tran_technology.cpp`); in the XML, *I* is the `coefficient` of the fuel input, *N* the `input-cost` of the `minicam-non-energy-input` named "non-energy", and *L* the `loadFactor`. The non-fuel cost is built in gcamdata (`zenergy_L154.transportation_UCD.R`) from capital and operating costs: capital is annualised with a 10 % discount rate over 10 years and divided by the distance a vehicle travels each year.

$$\begin{aligned} N = [\ &(\text{purchase} + \text{other} + \text{infrastructure}) \times \text{CRF}(10\ \%, 10\ \text{y}) + \text{maintenance} + \\ &\text{registration and insurance} + \text{tolls}\ ] \div \text{annual vkm} \end{aligned} \tag{3.4}$$

SIMPLE CALCULATION

Worked example: a compact car in the Middle East, 2020 (2005$)

1. CRF = 0.10 × 1.1^{10} ÷ (1.1^{10} − 1) = 0.1627.
2. Petrol car capital: 31,342 purchase + 10,970 other = 42,312 $; × 0.1627 = 6,886 $/yr.
3. Operating: 448 maintenance + 1,550 registration and insurance = 1,998 $/yr.
4. Non-fuel per vehicle-km: (6,886 + 1,998) ÷ 14,000 km = 0.635 $/vkm.
5. Fuel: 3.32 MJ/vkm × 10 $/GJ (illustrative) = 0.0332 $/vkm.
6. Per passenger-km: (0.635 + 0.0332) ÷ 2.33 persons = **0.286 $/pkm**.
7. Battery car, same steps (capital 51,991 $ incl. 1,409 $ charger; 1.03 MJ/vkm at 25 $/GJ): **0.328 $/pkm**.

Answer: the BEV is 15 % dearer per pkm in 2020; capital is 74 % of the petrol car's cost and fuel only 5 %

Data: `UCD_trn_data_CORE.csv` (petrol car, operating costs, intensities, load factor 2.33, 14,000 vkm/yr) and `OTAQ_trn_data_EMF37.csv` (BEV purchase, other and charger costs), Middle East, Compact Car, 2020, v9.1. Fuel prices are illustrative round numbers, not GCAM outputs.

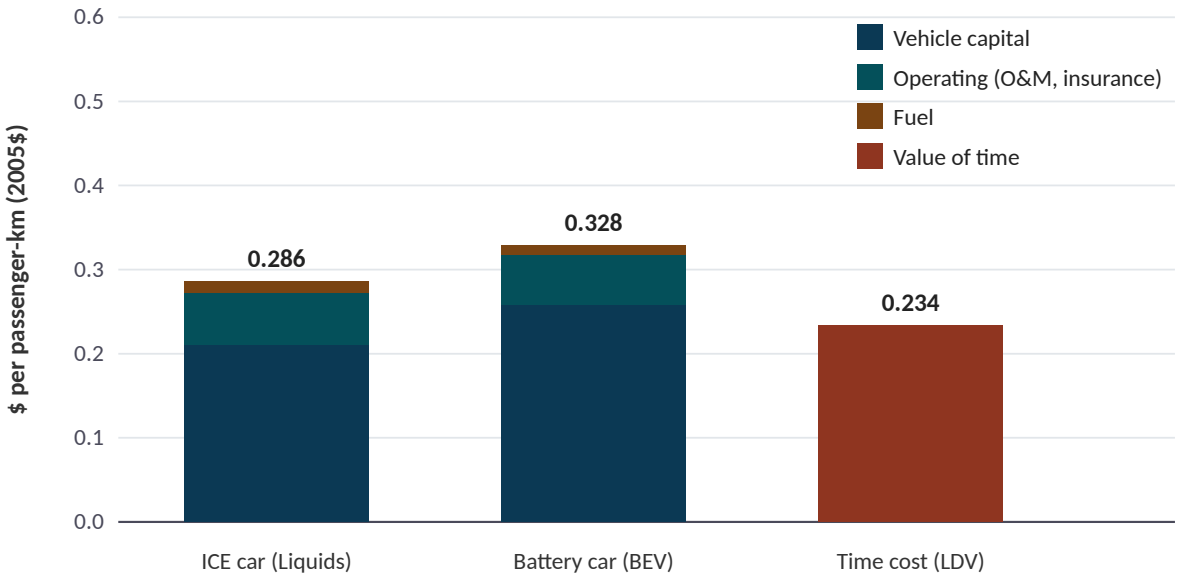


Figure 3.3 — Cost build-up per passenger-km for the worked example, and the value of time for a car trip at an illustrative GDP of $20,000 per person. What to notice: fuel is a thin slice; vehicle capital dominates, and the time cost is as large as the whole money cost.

RESEARCH NOTE

Which file sets the cost you think you are changing? For cars, trucks, buses, rail, aviation and ships, gcamdata overwrites the UC Davis (UCD) values with `OTAQ_trn_data_EMF37.csv`: "MODIFIED intensity and capital cost … based on AUTONOMIE and TEMPO harmonization" (source workbook dated 14 March 2023). In our example the petrol car's costs still come from UCD (no OTAQ row), but the BEV's come from OTAQ. Edit the wrong file and nothing changes.

Step 2: the generalised cost of a mode

A traveller also pays with time. GCAM converts income into an hourly wage, *W* = GDP per person ÷ 2,000 hours (40 hours × 50 weeks, hard-coded in `TranSubsector::getTimeValue`), scales it by a **time-value multiplier** *V* and divides by the mode's door-to-door **speed** *S*:

$$P_i = \Sigma_j \, s_{ji} \, P_{ji} + W \cdot V_i \div S_i, \quad W = \text{GDP per capita} \div (40\ \text{h} \times 50\ \text{wk}) \tag{3.5}$$

The time term is added only where `addTimeValue` = 1: on passenger modes and on the 4W node (once, at the LDV-versus-other-modes level), never on freight or on drivetrain choice. Since v8.5 the GDP used here is deflated from 1990 to 1975 dollars inside the code.

Mode	Time-value multiplier V (× wage)	Built from	Example speed, Middle East (km/h)
Domestic aviation	1.244	1.45 in-vehicle, 1.066 wait/walk, 53.7 % wait share	197
International aviation	1.291	41.5 % wait share	337
Rail, HSR	0.80	0.77 in-vehicle, 10.2 % wait	42 (HSR 170)
Bus	0.808	0.77 in-vehicle, 12.8 % wait	21–24
Cars, 2W/3W	0.82	no waiting	33 (2005) → 40 (2050)
Walk, Cycle	1.066	—	4–4.8 / 15

SIMPLE CALCULATION

Worked example: the price of time

1. GDP per person $20,000 (illustrative) ÷ 2,000 h = W = $10 per hour.
2. Car: 10 × 0.82 ÷ 35 km/h = 0.234 $/pkm.
3. Bus: 10 × 0.808 ÷ 21 km/h = 0.385 $/pkm — slower, so time costs more than the bus fare.
4. Rail: 10 × 0.80 ÷ 42 km/h = 0.190 $/pkm.
5. Double the income and every time cost doubles; fast modes gain on slow ones.

Answer: income growth pushes travellers towards faster modes even if money costs stay the same

Step 3: sharing — two kinds of logit

At every branch, shares follow a logit. GCAM has two forms (documentation, *Economic Choice*). The **relative-cost logit** compares cost *ratios*; the **absolute-cost logit** compares cost *differences*, scaled by a base value p_0:

$$\text{relative: } s_i = \alpha_i \, P_i^{\gamma} \div \Sigma_k \, \alpha_k \, P_k^{\gamma} \qquad \text{absolute: } s_i = \alpha_i \, e^{\gamma \, P_i/p_0} \div \Sigma_k \, \alpha_k \, e^{\gamma \, P_k/p_0} \tag{3.6}$$

Which form applies where? In the gcamdata files every tranSubsector (choice among drivetrains) is set to `absolute-cost-logit`, as is the `trn_pass_road_LDV_4W` sector. The other supplysectors leave `logit.type` empty, and gcamdata then fills in the default, the **relative-cost logit** (`add_logit_tables_xml` in `R/xml.R`). So **mode choice is relative, drivetrain choice is absolute**. The absolute form copes with costs that can be near zero or negative, which matters once carbon prices and subsidies enter.

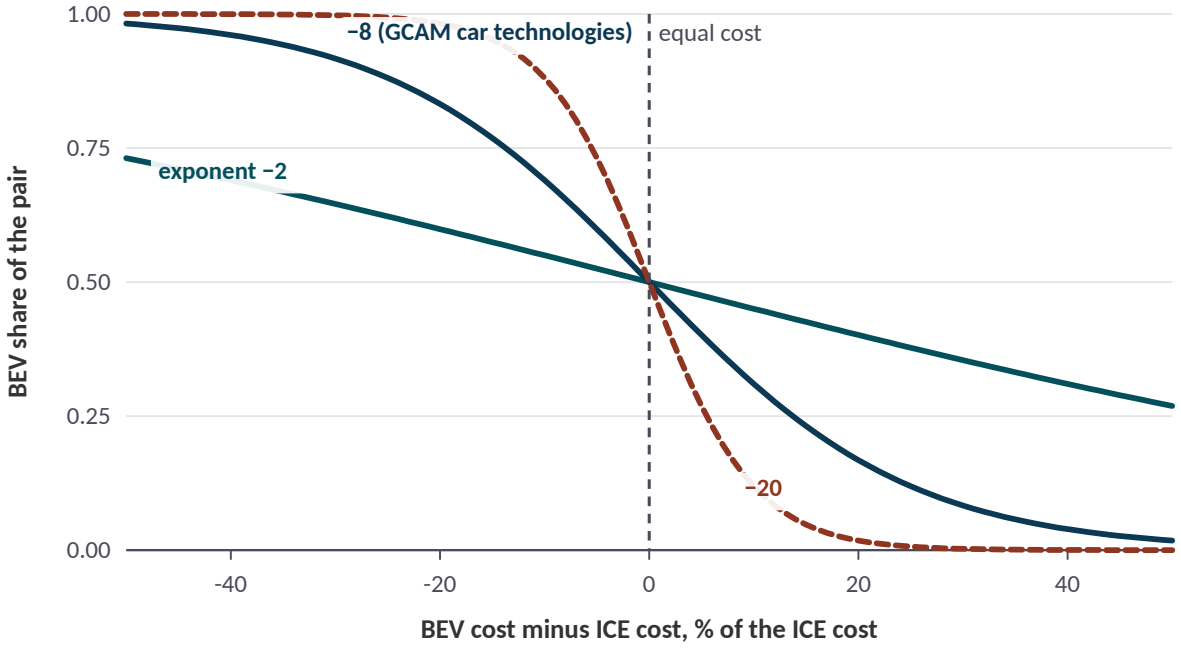


Figure 3.4 — BEV share of a two-way choice against its cost premium, absolute-cost logit with p_0 = ICE cost and equal share-weights. What to notice: with GCAM's −8 for cars, a 10 % premium leaves the BEV 31 % of the pair and a 10 % discount gives it 69 %.

SIMPLE CALCULATION

Worked example: calibrating a share-weight

1. Observed 2021: BEVs take 1 % of new car pkm, though they cost 15 % more ($p_{BEV} = 1.15\ p_0$).
2. With $\alpha_{ICE} = 1$ and $\gamma = -8$: $s_{BEV}/s_{ICE} = \alpha_{BEV} \cdot e^{-8(1.15\ -\ 1)} = \alpha_{BEV} \cdot e^{-1.2}$.
3. Set the ratio to 0.01/0.99 = 0.0101: $\alpha_{BEV} = 0.0101 \div 0.3012 = 0.0335$.
4. Check: share = 0.010, the observed 1 %.
5. Now the future path of α_{BEV} (0.034 → 1 by 2030 in the s-curve) does as much work as any cost change.

Answer: $\alpha_{BEV} \approx 0.034$; the share-weight hides everything costs do not explain

Step 4: how much travel

The total service demand grows with income and population and falls with the aggregate price of the service (the average cost that the nest passes up). In `minicam_price_elasticity_function.cpp`:

$$D_t = D_{t-1} \cdot (Y_t/Y_{t-1})^{\alpha} \cdot (P_t/P_{t-1})^{\beta} \cdot (N_t/N_{t-1}) \tag{3.7}$$

Y is GDP per person, P the service price, N population (only for per-capita demands); α and β are the income and price elasticities in the table above. Because α = 1 for passenger travel, travel per person grows as fast as income unless prices rise.

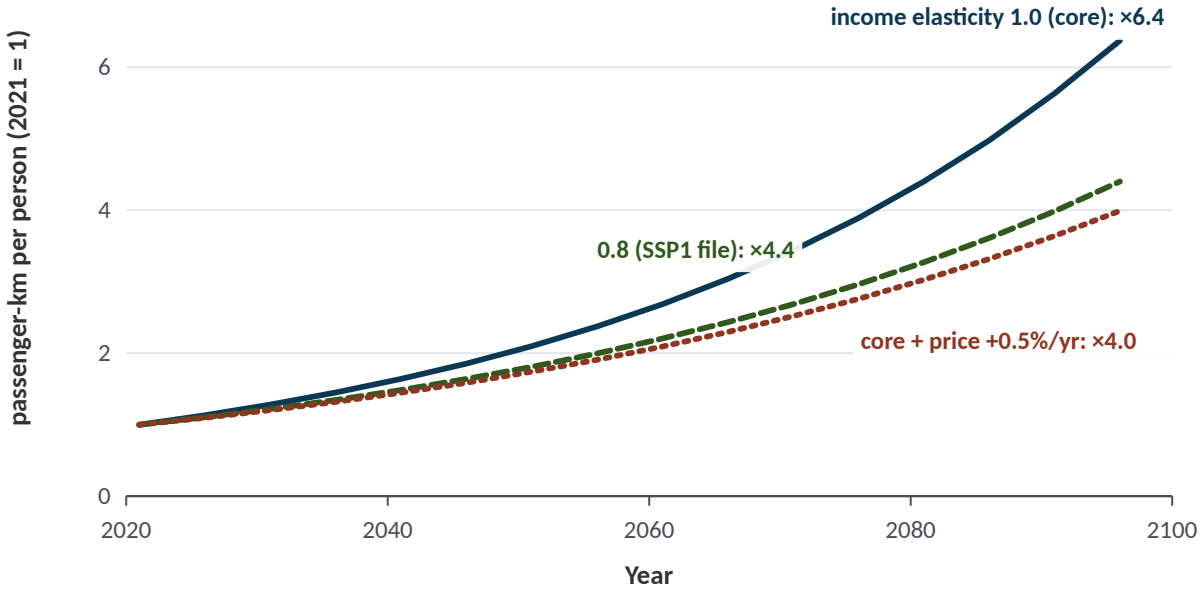


Figure 3.5 — Travel per person from Eq. 3.7 with income per person growing 2.5 %/yr (illustrative). What to notice: the SSP1 elasticity (0.8) cuts 2100 travel from ×6.4 to ×4.4; a price rising 0.5 %/yr does about as much, because the price elasticity is −1.25.

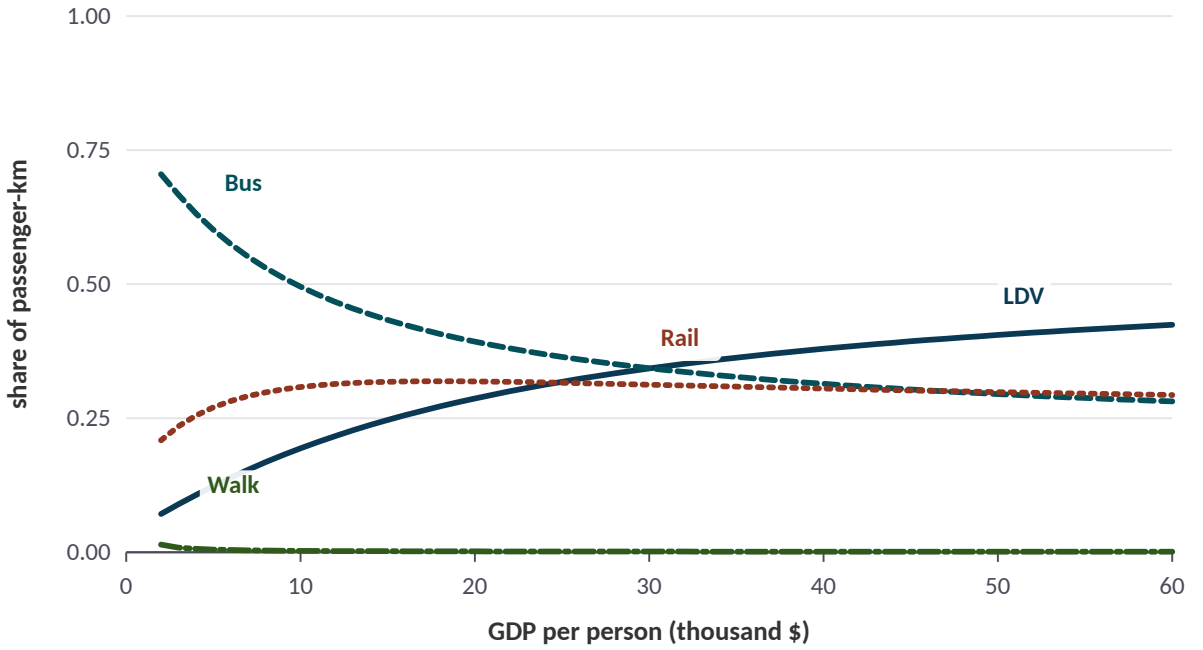


Figure 3.6 — Mode shares from Eq. 3.5 and a relative-cost logit (γ = −2) as income rises, with illustrative fares and share-weights. What to notice: the same money costs give a bus-dominated world when people are poor and a car-dominated one when they are rich, purely through the value of time.

IN GCAM

Where it lives (v9.1). Chunks: `input/gcamdata/R/zenergy_L154.transportation_UCD.R` (calibration, costs, intensities by UCD region) and `zenergy_L254.transportation_UCD.R` (level-2 tables such as `L254.StubTranTechCost`, `L254.StubTranTechLoadFactor`, `L254.tranSubsectorVOTT`, `L254.IncomeElasticity_trn`); XML writer `zenergy_xml_transportation_UCD_CORE.R` → `transportation_UCD_CORE.xml`, `_SSP1`, `_SSP3`, `_SSP5`. Assumptions: `A54.*_revised.csv` (the `_revised` files are the ones used), `UCD_trn_data_*.csv`, `OTAQ_trn_data_EMF37.csv`, mappings `UCD_techs_revised.csv`, `UCD_ctry.csv`.

Calibration. IEA transport energy by country and fuel is split into UCD modes, size classes and technologies with 2005 UCD shares, scaled to GCAM end-use energy, converted to vkm (÷ intensity) and pkm (× load factor); share-weights then reproduce each base year to 2021. Parameters exist for 14 UCD regions and are mapped to GCAM's 32: Saudi values are generic "Middle East" values.

Version	Transport change
v6.0 (2022)	CMP 359: hydrogen and transport update; bottom-up LDV battery and fuel-cell costs; BEV/FCEV trucks, buses, locomotives, ships, aircraft; OTAQ/EMF37 file added
v7.0 (2023)	gcamdata chunks renamed to `zenergy_L154…/L254…`
v7.1 (2024)	CMP 386: trucks re-characterised with Autonomie/BEAN; charger costs on vehicles; bus load factors halved in Africa, India, SE Asia
v8.0 / v8.1	final base year 2021 / Ukraine a separate region
v8.5 (2025)	units: billion pkm/tkm, 1975$, EJ per billion vkm (found by diffing the repository)
v8.8	hydrogen inputs renamed: `H2 LDV`, `H2 MHDV`, `LH2`; fuel-cell share-weights cut from 1 to 0.5
v9.0–9.1 (2026)	KLEAM macro module; transport capital tracked (LDV depreciation 1/15 per year)

TRAP

Two documentation pages disagree with the code. `en_technologies.html` still gives 1990$ and BTU per vehicle-km (v8.2 units). And the docs' general advice that passenger transport suits the relative-cost logit is true for mode choice but not for drivetrain choice, which the input files set to the absolute-cost form. Always check the XML you are running.

TRY IT NOW

1. Using Eq. 3.3, what happens to the petrol car's cost per pkm if the load factor falls from 2.33 to 2.0 (the 2050 value)? (Answer key.)
2. With $\gamma = -8$, how much cheaper must a BEV be than a petrol car for it to take 80 % of the pair, with equal share-weights?
3. Why does a carbon price of \$200/tCO_2 change car choice less than a 30 % change in battery-car capital cost? Use Figure 3.3.

RECAP SO FAR

1. **3.1 GCAM as research software:** data → gcamdata → XML → C++ core → database → queries; add-ons load in order; check the log before you trust a number.
2. **3.2 Transport in GCAM:** four final demands; cost per pkm = (fuel × intensity + non-fuel) ÷ load factor, plus wage × multiplier ÷ speed at mode level; modes share by relative-cost logit, technologies by absolute-cost logit.

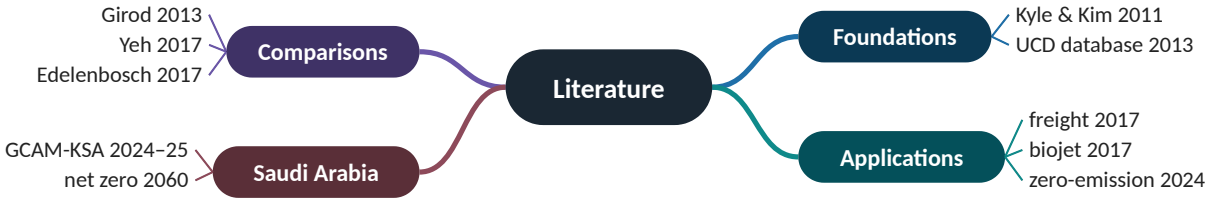
MODULE 3 OF 8

What the literature knows

More than thirty studies, grouped by theme: what is settled, what is disputed and what nobody has done yet.

HOOK

When eight global models were run with the same climate policy, almost all of the passenger-transport emission cuts came from cleaner fuels and more efficient vehicles; travelling less or switching to rail barely registered (Edelenbosch et al. 2017). Is that a fact about the world, or about how the models were built? That question runs through this whole module.



How this review was built

The list was compiled on 23 October 2026 from journal searches, PNNL/OSTI records, JGCRI model proposals and the KAPSARC publication pages. Each DOI below was checked against a publisher, PNNL, OSTI, IDEAS or Semantic Scholar record. Items whose DOI, pages or model list could not be confirmed are marked [U]. Read the papers themselves before citing a finding; the one-line summaries here are a map, not a substitute.

Theme A — Foundations: how GCAM transport was built

Study	Question	Model and change	Key finding
Kyle & Kim (2011) *Energy Policy* 39, 3012–3024. doi:10.1016/j.enpol.2011.03.016	Value of BEV, FCEV, NGV and advanced ICE	Early GCAM with detailed LDV competition [U]	Electric and H_2 vehicles lower 2095 CO_2 concentration by about 25 ppmv, more than advanced ICE or gas cars
Mishra et al. (2013) UC Davis ITS report UCD-ITS-RR-13-05 (grey)	Build the transport database	14 UCD regions, 2005 base, all modes and size classes	Source of today's `UCD_trn_data_*` files; still the main description of the database
Calvin et al. (2019) *GMD* 12, 677–698. doi:10.5194/gmd-12-677-2019	Document GCAM v5.1	Reference paper	Describes the nesting and time value; the paper to cite for structure
Bond-Lamberty et al. (2019) *J. Open Res. Softw.* doi:10.5334/jors.232 [U]	The data system	gcamdata R package	Chunk architecture used in module 3.5
Binsted et al. (2022) *GMD* 15, 2533–2559. doi:10.5194/gmd-15-2533-2022 [U]	Subnational US model	GCAM-USA	Template for a country-detail variant such as GCAM-KSA
Kyle et al. (2022) CMP 359; Kyle et al. (2024) CMP 386, PNNL-35913 (grey)	Update H_2 and vehicle data	v6.0 and v7.1 core changes	The technology data now in the release

Theme B — Model comparisons and realism

Study	Question	Model and change	Key finding
Girod et al. (2013) *Climatic Change* 118, 595–608. doi:10.1007/s10584-012-0663-6	Compare transport in five global models	GCAM among them (authorship) [U]	A \$200/tCO_2 tax in 2050 stops or reverses transport emission growth in all models
Pietzcker et al. (2014) *Energy* 64, 95–108. doi:10.1016/j.energy.2013.08.059	Transport demand and climate policy	Five energy-economy models [U]	Transport lags other sectors by 10–30 years at a given carbon price but can exceed 90 % cuts late in the century
Creutzig et al. (2015) *Science* 350, 911–912. doi:10.1126/science.aac8033	Is transport a roadblock?	Perspective	IAMs under-represent demand-side and behavioural options
Edelenbosch et al. (2017) *Transp. Res. D* 55, 281–293. doi:10.1016/j.trd.2016.07.003	Decompose passenger futures	About 8 IAMs incl. GCAM; Laspeyres decomposition	Cuts come from intensity and fuel switching; mode shift and activity contribute little
Yeh et al. (2017) *Transp. Res. D* 55, 294–309. doi:10.1016/j.trd.2016.11.001	Compare model structures and projections	GCAM, MESSAGE-Transport, IEA MoMo, ICCT Roadmap [U]	Large base-year data differences; IAMs favour low-carbon fuels, sectoral models favour efficiency
McCollum et al. (2017) *Transp. Res. D* 55, 322–342. doi:10.1016/j.trd.2016.04.003	Behavioural realism in vehicle choice	MESSAGE-Transport with consumer groups and disutility costs	Preferences slow alternative vehicles unless non-price policies are added
McCollum et al. (2018) *Nature Energy* 3, 664–673. doi:10.1038/s41560-018-0195-z	Preferences and climate policy	Multi-model (not GCAM)	Carbon pricing alone does not bring low-carbon vehicles to the mass market
Muratori et al. (2020) *Renew. Sustain. Energy Rev.* 119, 109541. doi:10.1016/j.rser.2019.109541	Integrated mobility–energy modelling	Review	Calls for coupled models with more spatial and temporal detail
Rottoli et al. (2021) *Transp. Res. D* 99, 103005. doi:10.1016/j.trd.2021.103005	Direct vs indirect electrification of cars	REMIND/EDGE-T (a comparator) [U]	Fuel cells or e-fuels need nearly twice the primary energy of BEVs

Theme C — GCAM applications: modes, fuels, countries

Study	Question	Model and change	Key finding
Chaturvedi & Kim (2015) *Energy Policy* 81, 176–185. doi:10.1016/j.enpol.2014.11.013	Global shift to electric rail	GCAM [U]	Transport emissions fall 8–49 %; carbon prices 3–17 % lower
Yin et al. (2015) *Energy Policy* 82, 233–248. doi:10.1016/j.enpol.2015.03.021	China transport in a global frame	Detailed China transport in GCAM	Carbon prices cut emissions but barely shift demand or modes
Muratori et al. (2017) *Environ. Sci. Technol.* 51, 3526–3533. doi:10.1021/acs.est.6b04515	Role of freight	GCAM	Freight grows fastest; mitigation relies on low-carbon liquids and demand reduction
Wise et al. (2017) *Transp. Res. D* 52, 244–253. doi:10.1016/j.trd.2017.03.006 [U]	Biojet in aviation	GCAM with biojet pathways	Biojet, especially with CCS, decarbonises aviation without disrupting other bioenergy
Paladugula et al. (2018) *Energy Policy* 116, 10–18. doi:10.1016/j.enpol.2018.01.037 [U]	India transport to 2050	Five teams [U]	Reference transport CO_2 grows about sevenfold 2010–2050
Ou et al. (2021) *Applied Energy* 300, 117364. doi:10.1016/j.apenergy.2021.117364	Large-scale US EV deployment	GCAM-USA	High EV adoption lowers net US CO_2 to 2050 even with a fossil-heavy grid
Wolfram et al. (2022) *Nature Energy* 7, 1112–1114. doi:10.1038/s41560-022-01124-4	Ammonia as a shipping fuel	GCAM with an ammonia extension [U]	Unmanaged NH_3/N_2O releases could perturb the nitrogen cycle
Browning et al. (2023) *Energy Clim. Change* 4, 100104. doi:10.1016/j.egycc.2023.100104	US net zero 2050 (EMF 37)	Multi-model incl. GCAM-USA	Deep transport electrification plus clean power; origin of the EMF37 vehicle file
Speizer et al. (2024) *Nature Comms* 15, 4439. doi:10.1038/s41467-024-48424-9	A zero-emission global transport sector	GCAM 6.0 + DAC-based e-fuels, FT bio-jet/marine, H_2 aviation and ships	Zero direct emissions by 2050 feasible in the High case; aviation and shipping ≈ 34 % of savings; long-haul aviation break-even above \$1,300/tCO_2
Wolfram et al. (2024) *One Earth* 7, 885–895. doi:10.1016/j.oneear.2024.04.012	Value of a hydrogen economy	GCAM with CMP 386 hydrogen	Hydrogen lowers mitigation costs by up to 22 % by 2050
Sadavarte et al. (2025) *PLOS Climate* 4, e0000714. doi:10.1371/journal.pclm.0000714	BEV vs ICE climate and air-pollution footprints	GCAM-USA v7.0; BEV share-weights set to 1	BEV damages 2–3.5× lower; cumulative CO_2 break-even about 2 years
Muratori et al. (2021) *Prog. Energy* 3, 022002. doi:10.1088/2516-1083/abe0ad [U]; update Muratori et al. (2025) *Nat. Rev. Clean Technol.* 1, 827–845. doi:10.1038/s44359-025-00108-3	Status of EVs	Review	EVs were 22 % of global car sales in 2024: data for 2020–2025 share-weights

Theme D — Saudi Arabia and downscaling

Study	Question	Model and change	Key finding
Kamboj et al. (2024) KAPSARC Discussion Paper, doi:10.30573/KS--2024-DP01 [U]	Saudi transport to net zero 2060	GCAM-KSA v1.0 (GCAM 6.0, Saudi Arabia split from Middle East, 2015 calibration)	Transport CO_2 in 2060: 418 Mt (No Policy), 320 (Current), 158 (NDC continued), 115 (NZ2060), vs 176 Mt in 2015; ≈ 80 % of new 4W and 65 % of new truck sales zero/low-emission
Kamboj et al. (2024) *Energy Strategy Rev.* 55, 101537. doi:10.1016/j.esr.2024.101537	Economy-wide path to 2060	GCAM-KSA	NDC ambition must roughly triple; transport and industry hardest; carbon removal needed
Apeaning et al. (2025) *Energy Clim. Change* 6, 100184. doi:10.1016/j.egycc.2025.100184	Cost of delay	GCAM-KSA	Early action cuts net-zero policy costs by 38–72 %
Qiu et al. (2024) *Environ. Res. Lett.* 19, 064042 [U]	Direct air capture timing	GCAM-KSA	DACCS removes 0.28–0.33 $GtCO_2$/yr by 2060; early deployment saves about $420 bn
KAPSARC (2025) *Decarbonizing Road Freight Transportation in Saudi Arabia* (grey)	Freight to 2060	GCAM-KSA	Land freight CO_2 29.4 Mt in 2060 under net zero; rail 30.6 %, ZEVs 25.9 % of land freight
Sferra et al. (2026) *GMD* 19, 3157–3191. doi:10.5194/gmd-19-3157-2026	Downscale regional pathways to countries	DSCALE v0.1, 143 countries, MESSAGE/NGFS	Blends national trends and regional paths with a convergence year; splits final energy by sector

Synthesis: known, disputed, unknown

Known (several studies agree)	Disputed (results depend on model or assumption)	Unknown or barely studied
• Transport cuts lag power and buildings at the same carbon price • In IAMs, cuts come mostly from vehicle efficiency and fuel switching • Freight and aviation are the hardest residuals • Battery cars are the cheapest route for light-duty road transport	• How much travel and mode shift policy can deliver (models vs sectoral studies) • Role of hydrogen and e-fuels vs biofuels in aviation and shipping • Speed of BEV uptake: share-weights and preferences decide it • Size of the time-value effect	• GCAM transport studies for Latin America since 2023 (none found) • Explicit vehicle stocks inside GCAM • Oil-exporter fuel-price reform and transport demand (Gulf hindcasts) • Sub-national charging load from global scenarios

RESEARCH NOTE

Reading a model-comparison paper. Ask four questions: (1) Were the scenarios harmonised in GDP, population and policy, or only in name? (2) Which differences come from base-year data (Yeh et al. 2017 found these large) and which from structure? (3) Does the decomposition method (Laspeyres, LMDI) leave a residual? (4) Which options could a model not choose because it does not contain them? The last one is why "little mode shift" can be a structural result rather than an empirical one.

TRAP

Citing a finding without its version. Kyle & Kim (2011) used a GCAM with different technologies, costs and share-weight paths from today's v9.1; the GCAM-KSA papers use GCAM 6.0 with a 2015 base year. A finding is a statement about a model version under stated assumptions. Report it that way, and do not compare numbers across versions without saying so.

TRY IT NOW

1. Pick one study from Theme C. Write its research question as a single decision someone could take, and name the output variable that answers it.
2. Which two studies would you put side by side to argue that GCAM under-states mode shift? What evidence would change your mind?

RECAP SO FAR

1. **3.1 GCAM as research software:** data → gcamdata → XML → C++ core → database → queries; add-ons load in order; check the log before you trust a number.
2. **3.2 Transport in GCAM:** four final demands; cost per pkm = (fuel × intensity + non-fuel) ÷ load factor, plus wage × multiplier ÷ speed at mode level; modes share by relative-cost logit, technologies by absolute-cost logit.
3. **3.3 Literature:** IAM transport mitigation comes mostly from intensity and fuel switching; demand and behaviour are under-represented; GCAM-KSA puts Saudi transport at 115 Mt in 2060 under net zero.

MODULE 4 OF 8

Designing a study

A good GCAM study answers one question someone has to decide, with scenarios built to isolate the answer.

HOOK

Four uncertain inputs with three values each make 81 GCAM runs. Add three policies and you have 243. At two hours a run on four cores, that is about five days of computing — before a single one has failed to solve. Design is the step where you decide which runs answer the question and which only fill a hard drive.

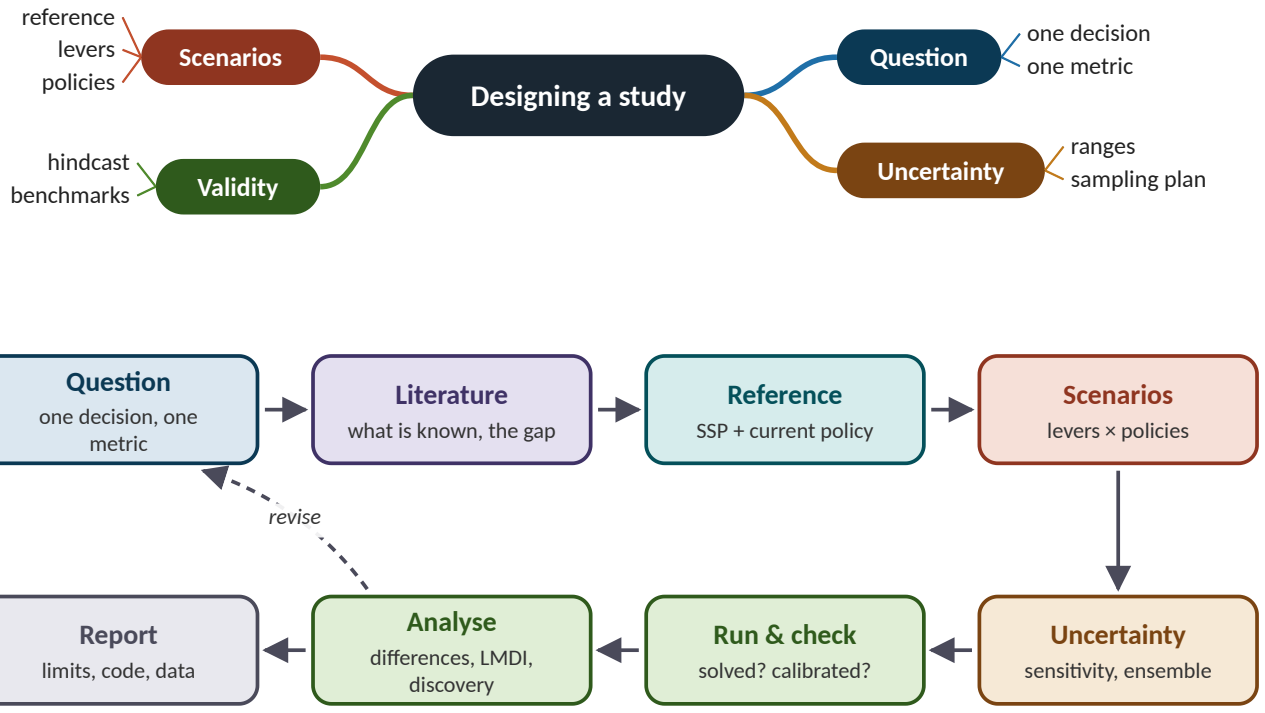


Figure 3.7 — The study design loop. What to notice: analysis feeds back into the question. Most good studies sharpen their question after the first round of runs.

From topic to question

"EVs in Saudi Arabia" is a topic. A research question names a **decision**, a **metric** and a **comparison**:

Weak	Strong	Why stronger
What is the future of EVs in KSA?	How much do 2060 Saudi transport CO_2 emissions differ between a ZEV sales standard and a carbon price that achieves the same 2040 car-emission cut, and at what cost?	Decision (instrument choice), metric (CO_2, cost), comparison (equal-effort policies)
Can aviation be decarbonised?	At what CO_2 price does bio-jet with CCS beat DAC offsets for Gulf aviation residuals, and how sensitive is it to biomass supply?	A threshold and a sensitivity, both computable
Will rail help freight?	How much do Saudi freight emissions fall by 2050 if rail's share-weight doubles, and how much of that survives a freight logit exponent of −4 instead of −2?	Separates the policy effect from a structural assumption

Scenario logic: Reference, levers and policies

Every GCAM study compares runs that differ in a controlled way. Three layers are enough for most transport questions:

1. **Background world**: an SSP (population, GDP, the matching transport file `transportation_UCD_SSP1/3/5.xml`; SSP2 uses CORE) and current policies.
2. **Levers**: the assumptions your question is about — BEV cost, share-weight path, load factors, speed of rail, a fuel technology you add.
3. **Policies**: none, a carbon price, an emissions constraint (net zero), a standard (ZEV credits, fuel quantity cap).

Hold everything else at Reference. If two things change at once, you cannot say which caused the difference.

A worked design matrix (from the kit)

Scenario	Technology lever	Behaviour lever	Policy	Purpose
Reference	Reference	Reference	None	baseline for all comparisons
BEVlow	BEV car non-fuel cost -30% from 2030	Reference	None	technology-cost sensitivity
LFup	Reference	car load factor +20% by 2050	None	demand-side behaviour
ZEVpref	faster BEV/FCEV share-weights	Reference	None	non-cost barriers fall (assumption)
TruckEff	heavy diesel truck intensity -25% by 2050	Reference	None	freight efficiency standard proxy
CTax	Reference	Reference	carbon tax 25 $/tC 2025 +5%/yr	price instrument
NetZero	Reference	Reference	regional CO2 cap to net zero 2060	quantity instrument (solved price)
ZEVstd	Reference	Reference	ZEV share of car pkm 10%->70% (2030-2050)	sector standard
BEVlow_CTax	BEV car non-fuel cost -30% from 2030	Reference	carbon tax 25 $/tC 2025 +5%/yr	interaction of technology and policy
BEVlow_NetZero	BEV car non-fuel cost -30% from 2030	Reference	regional CO2 cap to net zero 2060	does cheap BEV lower the carbon price?
Ref_SSP1	Reference	Reference	None	socioeconomic sensitivity
Ref_SSP3	Reference	Reference	None	socioeconomic sensitivity

From `go3-transport-kit/templates/scenario_design_matrix.csv`.

SIMPLE CALCULATION

How many runs, and how long?

1. Full factorial: 3 SSPs × 2 BEV cost levels × 2 share-weight paths × 3 policies = 36 runs.
2. At about 2 hours per run (machine-dependent; check your own) and 4 runs in parallel: 36 × 2 ÷ 4 = 18 hours.
3. Add a 3-level freight exponent test only for the central SSP and two policies: + 3 × 2 = 6 runs.
4. Allow 10 % for failed runs and re-runs: about 46 runs in total.

Answer: ≈ 46 runs, a day of computing — feasible; the 243-run version is not

Policies: price or quantity?

GCAM implements a carbon price (`fixedTax`), an emissions constraint (`constraint`, for which the model solves the shadow price), and portfolio standards (credit markets). A price fixes cost and lets emissions follow; a

cap fixes emissions and lets the price follow. For comparisons of instruments, match them on one dimension: either equal emissions (then compare cost) or equal price (then compare emissions). Go 2 showed that a cap and a price can be two doors to the same room; in a study you must say which door you used.

Instrument	You specify	GCAM returns	Good for
Carbon tax	price path ($/tC, 1990$)	emissions, technology mix	"what does $X buy?"
Emissions constraint	emissions path	the shadow price	net-zero targets, budgets
Policy portfolio standard	a credit market and its constraint	credit price, compliance mix	ZEV mandates, fuel standards
Share-weight or cost change	an assumption	uptake	"what if technology X were cheaper/preferred?" (not a policy!)

TRAP

Calling an assumption a policy. Setting BEV share-weights to 1 makes BEVs popular; it does not say which policy would make them popular, or what it would cost. If your scenario is "high EV preference", name it as an assumption and, if you can, pair it with a policy (credit standard, subsidy via `input-subsidy`) whose cost GCAM reports.

Plan the uncertainty before you run

Write a table of 3–6 uncertain inputs, each with a low, central and high value *and a source*. Choose a design: one-at-a-time (cheap, misses interactions), full factorial (clean, explodes in size), or a Latin-hypercube sample (efficient for many inputs, analysed with regression or scenario discovery). Module 3.6 analyses the results.

Uncertain input	Low	Central	High	Source of the range
BEV non-energy cost (2030)	−30 %	release value	+30 %	BNEF pack prices ($108/kWh in 2025); NREL ATB
BEV share-weight path	v5.4-style (0.25 in 2030)	release s-curve (1 by 2030)	1 from 2025	repository history; IEA EV sales (22 % in 2024)
Drivetrain logit exponent	−4	−8	−12	sensitivity convention
Passenger income elasticity	0.8 (SSP1 file)	1.0	1.1	A54 files; literature
Load factor 2050	1.8	2.0	2.3	UCD trend vs 2020 level

Validity: why should anyone believe it?

- **Calibration check**: base-year outputs reproduce IEA energy by fuel (they should, by construction — so this checks the data, not the model).
- **Hindcast**: run from 2015 with observed prices and compare 2021–2024 with data you did not calibrate to (e.g. Saudi gasoline demand after the 2016 and 2018 price reforms; IEA EV sales).
- **Benchmark**: compare your Reference with other models in the AR6 database or EMF studies for the same region and variables.

- **Face validity**: show results to people who know the sector; implausible numbers usually reveal an input error.

RESEARCH NOTE

Pre-register your design. Before the first run, write down the scenarios, the uncertain ranges and the figures you will make (the kit's `templates/research_proposal.md` has the headings). It protects you from running until something looks interesting, and it makes the honest "what the model cannot tell us" section much easier to write.

TRY IT NOW

1. Rewrite this into a strong question: "Hydrogen trucks in the Gulf". Name the decision, metric and comparison.
2. Your design has 2 SSPs × 3 BEV costs × 2 exponents × 3 policies. How many runs? Which factor would you drop first, and why?

RECAP SO FAR

1. **3.1 GCAM as research software:** data → gcamdata → XML → C++ core → database → queries; add-ons load in order; check the log before you trust a number.
2. **3.2 Transport in GCAM:** four final demands; cost per pkm = (fuel × intensity + non-fuel) ÷ load factor, plus wage × multiplier ÷ speed at mode level; modes share by relative-cost logit, technologies by absolute-cost logit.
3. **3.3 Literature:** IAM transport mitigation comes mostly from intensity and fuel switching; demand and behaviour are under-represented; GCAM-KSA puts Saudi transport at 115 Mt in 2060 under net zero.
4. **3.4 Study design:** one question, one decision metric; Reference + levers × policies; plan uncertainty before running.

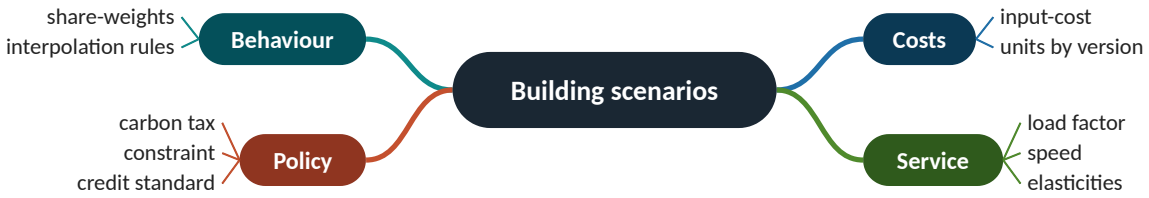
MODULE 5 OF 8

Building transport scenarios

Every scenario is a short XML file read after the core files. Small files, big consequences: get the names, units and order right.

HOOK

In GCAM's Middle East data, a petrol compact car costs about **$0.286** per passenger-km in money and a battery car **$0.328** (2020, our worked example). That 15 % gap gives battery cars only 23 % of the pair at the car exponent of −8. Cut the battery car's non-fuel cost by 30 % and the gap flips to -19 %, and its share jumps to 81 %. One number in one add-on file moves the whole story — which is why every number in that file needs a source.



The anatomy of an add-on file

An add-on XML file repeats the path from `<world>` down to the object you want to change and sets only the values you change. GCAM reads files in the order the configuration lists them; when a later file names an object that already exists, it **updates** it rather than creating a new one. So an add-on must come **after** the core file that creates the object, and every name on the path (region, supplysector, tranSubsector, technology) must match exactly. A misspelt name does not cause an error: it quietly creates a new, empty object that nothing uses.

BUILD IT · CONFIGURATION ORDER

```
<Configuration>
  <ScenarioComponents>
    <Value name = "transportation">../input/gcamdata/xml/transportation_UCD_CORE.xml</Value>

    <Value name = "trn_bev_cost_low">../input/user/trn_bev_cost_low.xml</Value>


    <Value name = "solver">../input/solution/cal_broyden_config.xml</Value>
  </ScenarioComponents>
  <Strings>
    <Value name="scenarioName">Trn_BEV_low</Value>
  </Strings>
</Configuration>
```

Path level	XML element	Transport example (v9.1)	Where the names come from
Region	`region name=`	Middle East, USA, China	`common/GCAM_region_names.csv`
Sector	`supplysector name=`	trn_pass_road_LDV_4W, trn_freight_road	`energy/A54.sector.csv`
Mode	`tranSubsector name=`	Car, Large Car and Truck, Bus, Passenger Rail	`A54.tranSubsector_logit_revised.csv`
Technology (regional)	`stub-technology name=`	Liquids, BEV, FCEV, NG, Hybrid Liquids	`mappings/UCD_techs_revised.csv`
Technology (global)	`tranTechnology` in `global-technology-database`	share-weights and interpolation rules	`A54.globaltranTech_shrwt_revised.csv`

IN GCAM

Element names were checked against the C++ XML names in `cvs/objects/` and the ModelInterface headers in `input/gcamdata/inst/extdata/mi_headers/ModelInterface_headers.txt` (gcam-core v9.1). The fastest way to get a correct path is to open the core file you are modifying, `transportation_UCD_CORE.xml`, find the object, and copy its parents.

Lever 1: costs

The non-fuel cost of a transport technology is its `input-cost` on the `minicam-non-energy-input`, in dollars per **vehicle**-km. GCAM divides it by the load factor to get dollars per passenger-km (Eq. 3.3). Fuel enters through a `minicam-energy-input` with a **coefficient** (energy per vehicle-km) whose price GCAM finds in the fuel market.

BUILD IT · CHEAPER BATTERY CARS

```
<scenario>
  <world>
    <region name="Middle East">
      <supplysector name="trn_pass_road_LDV_4W">
        <tranSubsector name="Car">
          <stub-technology name="BEV">
            <period year="2030">
              <minicam-non-energy-input name="non-energy">
                <input-cost>0.0700</input-cost>
              </minicam-non-energy-input>
            </period>
            <period year="2035">
              <minicam-non-energy-input name="non-energy">
                <input-cost>0.0680</input-cost>
              </minicam-non-energy-input>
            </period>
            <period year="2040">
              <minicam-non-energy-input name="non-energy">
                <input-cost>0.0660</input-cost>
              </minicam-non-energy-input>
            </period>
            <period year="2045">
              <minicam-non-energy-input name="non-energy">
                <input-cost>0.0650</input-cost>
…
```

SIMPLE CALCULATION

What a 30 % cut in BEV non-fuel cost does (Middle East compact car, 2020 values, illustrative fuel prices)

1. Non-fuel cost per vkm: 0.741 → 0.741 × 0.7 = 0.518 $/vkm.
2. Add fuel 0.0257 $/vkm, divide by load factor 2.33: (0.518 + 0.0257) ÷ 2.33 = 0.233 $/pkm.
3. Compare the petrol car at 0.286 $/pkm: BEV is now -18.5 % (it was +14.8 %).
4. Absolute-cost logit (Eq. 3.6), exponent −8, p_0 = ICE cost as in Figure 3.4: BEV share of the pair 23 % → 81 %.

Answer: BEV share 23 % → 81 % of the two-car choice (money cost only; time cost is the same for both)

TRAP

Units changed in v8.5. Transport non-fuel costs moved from 1990$ to **1975$** per vkm and intensities from BTU per vkm to **EJ per billion vkm**; service is in **billion** pkm/tkm. A 1990$ value must be multiplied by about 0.47 (gcamdata's `gdp_deflator(1975, 1990)` = 27.80 ÷ 59.31, from the BEA deflator series in `R/pipeline-helpers.R`) before it goes into a v9.1 add-on. Copying a v8.2 tutorial value unchanged makes the car about twice as expensive, and the model will still solve.

Lever 2: share-weights and their interpolation rules

Share-weights (module 3.1) are where calibration lives, so they are also where hidden assumptions live. Future share-weights for transport technologies are set globally in the `global-technology-database` and moved between years by **interpolation rules**. If you set a share-weight for 2025 but leave the default rule in place, the

rule will re-impose its own path for later years. To make your value stick, add an `interpolation-rule` with a `fixed` function (or a path you choose) in the same file.

BUILD IT · SHARE-WEIGHT PATH

```
<scenario>
  <world>
    <global-technology-database>
      <location-info sector-name="trn_pass_road_LDV_4W" subsector-name="Car">
        <tranTechnology name="BEV">
          <period year="2025">
            <share-weight>0.5</share-weight>
          </period>
          <period year="2030">
            <share-weight>1</share-weight>
          </period>
          <interpolation-rule apply-to="share-weight" from-year="2030" to-year="2100">
            <interpolation-function name="fixed"/>
          </interpolation-rule>
        </tranTechnology>
        <tranTechnology name="FCEV">
          <period year="2030">
            <share-weight>0.25</share-weight>
          </period>
          <period year="2040">
            <share-weight>0.5</share-weight>
          </period>
          <period year="2050">
            <share-weight>1</share-weight>
…
```

RESEARCH NOTE

Report share-weights as a scenario dimension. In the current files the BEV share-weight reaches 1 by 2030 on an s-curve; in v5.4 it was 0.25 in 2030 and 0.625 in 2050. A study that sets BEV share-weights to 1 from 2025 (as some US studies do) is answering "what if consumers treated BEVs like petrol cars from now on?" — a legitimate question, but not a forecast and not a policy. In module 3.6's tornado chart the share-weight path is the second-largest influence, after cost.

Lever 3: how much service, at what speed

Three mode-level levers change how people travel rather than what they drive. The **load factor** (passengers per vehicle) divides every per-vehicle cost and intensity; the **speed** of a mode sets its time cost; the **income and price elasticities** of `trn_pass` set how much travel there is at all (module 3.2).

SIMPLE CALCULATION

Fewer people per car: the load factor falls from 2.33 (2020) to 2.0 (the UC Davis 2050 value, petrol compact car)

1. Money cost per vkm = non-fuel 0.635 + fuel 0.0332 = 0.668 $/vkm.
2. At 2.33 passengers: 0.668 ÷ 2.33 = 0.286 $/pkm.
3. At 2.0 passengers: 0.668 ÷ 2.0 = 0.334 $/pkm, +17 %.
4. Energy and CO_2 per pkm rise by the same 17 %, because intensity is also per vehicle-km.

Answer: +17 % on cost, energy and CO_2 per passenger-km

BUILD IT · LOAD FACTOR

```
<scenario>
  <world>
    <region name="Middle East">
      <supplysector name="trn_pass_road_LDV_4W">
        <tranSubsector name="Car">
          <stub-technology name="Liquids">
            <period year="2050">
              <loadFactor>1.80</loadFactor>
            </period>
          </stub-technology>
          <stub-technology name="Hybrid Liquids">
            <period year="2050">
              <loadFactor>1.80</loadFactor>
            </period>
          </stub-technology>
          <stub-technology name="NG">
            <period year="2050">
              <loadFactor>1.80</loadFactor>
…
```

IN GCAM

Input element `<loadFactor>`; output database name `load-factor` (`reporting/source/xml_db_outputter.cpp`), which is what the query "transport load factors" reads. Mode-level levers sit on `tranSubsector`: `<share-weight year=…>`, `<speed year=…>`, `<time-value-multiplier year=…>`. Demand elasticities sit on `<energy-final-demand name="trn_pass">` as `<income-elasticity year=…>` and `<price-elasticity year=…>`.

BUILD IT · TRUCK EFFICIENCY

```
<scenario>
  <world>
    <region name="Middle East">
      <supplysector name="trn_freight_road">
        <tranSubsector name="Heavy truck">
          <stub-technology name="Liquids">
            <period year="2030">
              <minicam-energy-input name="refined liquids enduse">
                <coefficient>0.0110</coefficient>
                <market-name>Middle East</market-name>
              </minicam-energy-input>
            </period>
            <period year="2040">
              <minicam-energy-input name="refined liquids enduse">
                <coefficient>0.0100</coefficient>
                <market-name>Middle East</market-name>
              </minicam-energy-input>
            </period>
…
```

Lever 4: policies

GCAM has three policy objects that cover almost every transport policy question. Module 3.4 explained when to use which; here is how each looks.

A carbon price

BUILD IT · CARBON TAX

```
<scenario>
  <world>
    <region name="Middle East">
      <ghgpolicy name="CO2">
        <market>Middle East</market>
        <fixedTax year="2025">25.00</fixedTax>
        <fixedTax year="2030">31.91</fixedTax>
        <fixedTax year="2035">40.72</fixedTax>
        <fixedTax year="2040">51.97</fixedTax>
        <fixedTax year="2045">66.33</fixedTax>
        <fixedTax year="2050">84.66</fixedTax>
        <fixedTax year="2055">108.05</fixedTax>
        <fixedTax year="2060">137.90</fixedTax>
        <fixedTax year="2065">176.00</fixedTax>
        <fixedTax year="2070">224.63</fixedTax>
        <fixedTax year="2075">286.68</fixedTax>
        <fixedTax year="2080">365.89</fixedTax>
        <fixedTax year="2085">466.98</fixedTax>
        <fixedTax year="2090">596.00</fixedTax>
        <fixedTax year="2095">760.66</fixedTax>
…
```

SIMPLE CALCULATION

Units of a carbon price in GCAM

1. Policy files declare `<price-unit>1990$/tC</price-unit>`: dollars of 1990 per tonne of **carbon**.
2. One tonne of carbon is 44 ÷ 12 = 3.667 tonnes of CO_2.
3. So \$100 per tonne CO_2 (in 1990\$) = 100 × 3.667 = \$367 per tonne C.
4. Convert today's dollars to 1990\$ with the GDP deflator first; say which year's dollars your paper reports.

Answer: \$100/tCO_2 = \$367/tC (both in 1990\$)

An emissions constraint (net zero)

With `<constraint year=…>` instead of `fixedTax`, GCAM treats the emissions cap as a market and solves for the price that clears it (the shadow price). Linked gases enter through `<linked-ghg-policy>` with `price-adjust` (global-warming potential) and `demand-adjust`; setting `demand-adjust` to 0 for `CO2_LUC` restricts the cap to fossil and industrial CO_2, as the core file `global_ffict.xml` does.

BUILD IT · NET-ZERO CONSTRAINT

```
<scenario>
  <world>
    <region name="Middle East">
      <ghgpolicy name="CO2">
        <market>Middle East</market>
        <constraint year="2025">600</constraint>
        <constraint year="2030">514.3</constraint>
        <constraint year="2035">428.6</constraint>
        <constraint year="2040">342.9</constraint>
        <constraint year="2045">257.1</constraint>
        <constraint year="2050">171.4</constraint>
        <constraint year="2055">85.7</constraint>
        <constraint year="2060">0</constraint>
        <constraint year="2065">0</constraint>
        <constraint year="2070">0</constraint>
        <constraint year="2075">0</constraint>
        <constraint year="2080">0</constraint>
        <constraint year="2085">0</constraint>
        <constraint year="2090">0</constraint>
        <constraint year="2095">0</constraint>
        <constraint year="2100">0</constraint>
      </ghgpolicy>
    </region>
  </world>
…
```

A ZEV standard as a credit market

A mandate such as "30 % of car passenger-km must be zero-emission" becomes a market for credits. Every car technology **demands** 0.3 credits per unit of output; BEVs and FCEVs **supply** 1 credit per unit; a `policy-portfolio-standard` of type `RES` makes demand equal supply. GCAM finds the credit price that does it: a tax on petrol cars and a subsidy to zero-emission cars at the same time. Note the share is in **passenger-km**, not vehicle sales, because GCAM has no vehicle counts.

BUILD IT · ZEV CREDIT STANDARD

```
<scenario>
  <world>
    <region name="Middle East">
      <policy-portfolio-standard name="ZEV_credit">
        <market>Middle East</market>
        <policyType>RES</policyType>
      </policy-portfolio-standard>
      <supplysector name="trn_pass_road_LDV_4W">
        <tranSubsector name="Car">
          <stub-technology name="Liquids">
            <period year="2030">
              <minicam-energy-input name="ZEV_credit">
                <coefficient>0.16</coefficient>
                <market-name>Middle East</market-name>
              </minicam-energy-input>
            </period>
            <period year="2035">
              <minicam-energy-input name="ZEV_credit">
                <coefficient>0.32</coefficient>
                <market-name>Middle East</market-name>
              </minicam-energy-input>
            </period>
            <period year="2040">
              <minicam-energy-input name="ZEV_credit">
                <coefficient>0.56</coefficient>
                <market-name>Middle East</market-name>
…
```

SIMPLE CALCULATION

Balance of a 30 % ZEV credit market

1. Credit demand = 0.3 × (all car pkm). Credit supply = 1 × (ZEV car pkm).
2. Market clears when 1 × ZEV pkm = 0.3 × all pkm, so ZEV pkm ÷ all pkm = 0.3.
3. If ZEVs already exceed 30 % without the policy, supply exceeds demand: the credit price falls to zero (the constraint does not bind).
4. If they fall short, the credit price rises until enough cars switch — that price is the policy's marginal cost per credit.

Answer: the credit price is the cost of the last unit of compliance

RESEARCH NOTE

Not yet verified in a running model: (1) whether a transport-only carbon price can be set by renaming the CO_2 object on transport technologies (`CO2_TRN`); (2) whether negative `output-ratio` values are accepted for an intensity-style standard; (3) a sustainable-aviation-fuel mandate, which needs an aviation-specific liquids pass-through sector because aviation draws on the generic `refined liquids enduse`. Test each on one region and one period before building a study on it.

Batch runs: the scenario matrix as a file

GCAM's batch mode runs every combination of the component sets listed in a batch file. Each `ComponentSet` is one dimension (for example "BEV cost" with Reference and Low); each `FileSet` inside it is one level and lists

the add-on files for it. Point `BatchFileName` in the configuration to the batch file and set `BatchMode` to 1.

BUILD IT · BATCH FILE

```
<BatchRunner xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="batch_runner.xsd">
  <ComponentSet name="technology">
    <FileSet name="Ref">
    </FileSet>
    <FileSet name="BEVlow">
      <Value name="trn_bev_cost_low">../input/user/trn_bev_cost_low.xml</Value>
    </FileSet>
    <FileSet name="LFup">
      <Value name="trn_loadfactor_up">../input/user/trn_loadfactor_up.xml</Value>
    </FileSet>
    <FileSet name="BEVlow-LFup">
      <Value name="trn_bev_cost_low">../input/user/trn_bev_cost_low.xml</Value>
      <Value name="trn_loadfactor_up">../input/user/trn_loadfactor_up.xml</Value>
    </FileSet>
  </ComponentSet>
  <ComponentSet name="policy">
    <FileSet name="_NoPolicy">
    </FileSet>
    <FileSet name="_CTax">
      <Value name="policy_ctax">../input/user/policy_carbon_tax.xml</Value>
    </FileSet>
    <FileSet name="_NetZero">
      <Value name="policy_netzero">../input/user/policy_co2_constraint_netzero.xml</Value>
    </FileSet>
  </ComponentSet>
</BatchRunner>
```

Systematic changes: go through gcamdata

For changes to many regions or years, edit the source tables and rebuild the XML rather than writing long add-ons. Remember that `OTAQ_trn_data_EMF37.csv` overrides the UC Davis table for most vehicle costs and intensities, so editing `UCD_trn_data_CORE.csv` alone may change nothing. For a clean record of your change, write a **user-modification chunk** and pass it to the driver:

```
usermod_trn_ev <- function(command, ...) {
  if (command == driver.DECLARE_MODIFY) {
    return(c(FILE = "energy/A54.globaltranTech_shrwt_revised"))
  } else if (command == driver.DECLARE_INPUTS) {
    return(NULL)
  } else if (command == driver.MAKE) {
    all_data <- list(...)[[1]]
    x <- get_data(all_data, "energy/A54.globaltranTech_shrwt_revised")
    x[x$tranTechnology == "BEV" & x$supplysector == "trn_pass_road_LDV_4W", "2030"] <- 1
    return_modified("energy/A54.globaltranTech_shrwt_revised" = x)
  } else stop("Unknown command")
}
driver_drake(user_modifications = "usermod_trn_ev", xml_suffix = "__ev")
```

Pattern from `input/gcamdata/vignettes/usermod_vignette.Rmd` and `R/driver.R` (gcam-core v9.1). Check how the year column is named after reading ("2030" vs `X2030`) in your version.

TRAP

"My add-on changed nothing." The usual causes, in order: (1) the file is listed before the core transport file; (2) a name on the path is misspelt or uses a v8.2 name (hydrogen fuels were renamed in v8.8 to `H2 LDV`, `H2 MHDV`, `LH2`); (3) an interpolation rule overwrote your share-weight; (4) the value is in the old units and so is a factor of 1,000 off in the other direction. Always query the variable you changed.

TRY IT NOW

1. Write the XML path (just the element names and attributes) that sets the speed of Passenger Rail in the Middle East to 60 km/h in 2040.
2. A v8.2 paper reports a BEV non-fuel cost of 0.30 1990$/vkm. What value goes into a v9.1 add-on? (Use the deflator ratio above.)
3. Design a ZEV credit market for trucks only. Which technologies demand credits, which supply them, and what is the coefficient for a 20 % standard?

RECAP SO FAR

1. **3.1 GCAM as research software:** data → gcamdata → XML → C++ core → database → queries; add-ons load in order; check the log before you trust a number.
2. **3.2 Transport in GCAM:** four final demands; cost per pkm = (fuel × intensity + non-fuel) ÷ load factor, plus wage × multiplier ÷ speed at mode level; modes share by relative-cost logit, technologies by absolute-cost logit.
3. **3.3 Literature:** IAM transport mitigation comes mostly from intensity and fuel switching; demand and behaviour are under-represented; GCAM-KSA puts Saudi transport at 115 Mt in 2060 under net zero.
4. **3.4 Study design:** one question, one decision metric; Reference + levers × policies; plan uncertainty before running.
5. **3.5 Building scenarios:** add-on XML for costs, load factors, share-weights (with interpolation rules) and policies; mind the v8.5 units.

MODULE 6 OF 8

Analysing and stress-testing results

A result is a difference from Reference that survives the uncertainty you can name.

HOOK

In the kit's toy transport model, filling cars with more people (the LF-up scenario) should cut emissions. By 2050 it cuts them by only 3 Mt out of 84 — because cheaper car travel makes people travel 15 % more and pulls them out of buses. The first number you look at is rarely the answer; the analysis is in explaining it.



Step 1: get the numbers out, the same way every time

Write one script that runs every query you need for every scenario and saves one tidy table (scenario, region, year, variable, item, value, unit). Never copy numbers from ModelInterface by hand. The transport queries live in `output/queries/Main_queries.xml`; the kit's query file collects the ones a transport study needs.

Query title (Main_queries.xml, v9.1)	Line	Use
transport service output by mode	3023	pkm/tkm by mode (axis: subsector)
transport service output by tech (new)	3048	new-vintage shares: the closest thing to sales
transport final energy by fuel	2963	energy by carrier
transport load factors / transport speeds	3056 / 3198	check your levers took effect
costs of transport techs / modes	3084 / 3077	explain choices
CO2 emissions by sector (excluding resource production)	4532	direct CO_2
CO2 emissions by tech	4883	CO_2 by vehicle technology

BUILD IT · READ ALL SCENARIOS (PYTHON)

```
def from_db(db, dbname):
    import gcamreader
    conn = gcamreader.LocalDBConn(db, dbname)
    queries = gcamreader.parse_batch_query(QFILE)
    out = {}
    for key, title in WANT.items():
        q = [x for x in queries if getattr(x, "title", None) == title]
        if not q:
            raise SystemExit(f"query '{title}' not found in {QFILE}")
        out[key] = tidy(conn.runQuery(q[0]))
    return out


def from_sample():
    return {k: tidy(pd.read_csv(os.path.join(HERE, "sample_data", f))) for k, f in
SAMPLE.items()}

…
```

BUILD IT · READ ALL SCENARIOS (R)

```
  library(rgcam)
  conn <- localDBConn(args[2], ifelse(length(args) >= 3, args[3], "database_basexdb"))
  prj_file <- file.path(here, "out", "transport.dat"); dir.create(dirname(prj_file),
showWarnings = FALSE)
  scen <- listScenariosInDB(conn)$name
  for (s in scen) prj <- addScenario(conn, prj_file, s, qfile)
  data <- lapply(want, function(t) tidy(getQuery(prj, t)))
}
dir.create(file.path(here, "out"), showWarnings = FALSE)
for (k in names(data)) {
  p <- file.path(here, "out", paste0(k, "_R.csv")); write_csv(data[[k]], p)
  cat(sprintf("%-22s %5d rows, %d scenarios -> %s\n", k, nrow(data[[k]]),
length(unique(data[[k]]$scenario)), p))
}
```

Step 2: compare against Reference, then explain

The policy effect is the difference between two runs that share everything else. Report absolute differences (Mt) and relative ones (% of Reference), per capita where regions differ in size, and cumulatively where the climate is concerned.

Toy scenario	Year	pkm (bn)	car share	BEV share of car pkm	CO_2 (Mt)
Reference	2050	2,236	79 %	47 %	84
Reference	2100	3,523	89 %	47 %	142
BEV-low	2050	2,681	86 %	95 %	26
BEV-low	2100	3,953	91 %	95 %	39
Carbon-price	2050	2,198	79 %	54 %	64
Carbon-price	2100	3,398	93 %	74 %	11
Fast-wage	2050	2,488	83 %	47 %	97
Fast-wage	2100	3,727	91 %	47 %	152
LF-up	2050	2,581	85 %	45 %	81
LF-up	2100	3,865	91 %	45 %	127

From `go3-transport-kit/toy/gcam_transport_toy.py`: one illustrative region, GCAM-like nesting, logit types and exponents, value of time; no vintages, no fuel-price feedback. Python and R versions print the same table.

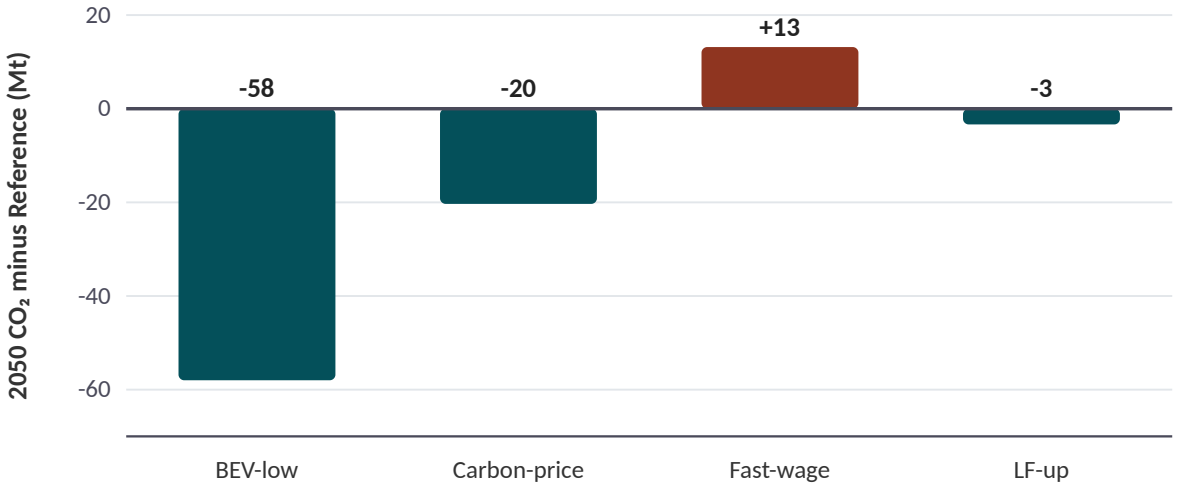


Figure 3.8 — 2050 CO_2 difference from Reference in the toy model. What to notice: cheaper battery cars cut 58 Mt; higher car occupancy cuts very little because it also makes car travel cheaper; faster wage growth raises emissions through more and faster travel.

RESEARCH NOTE

Two toy artefacts worth noticing. (1) In the toy Reference, the BEV share jumps to about half of car pkm in 2030, because the share-weight switches from 0 to 1 between 2025 and 2030 and the toy has no vintages. GCAM smooths this with an s-curve and capital stock. (2) Sales-like shares ("tech (new)") move much faster than the stock of vehicles on the road (Figure 3.15 in module 3.7). Always say which one you report.

Step 3: decompose the change

To explain *why* transport CO_2 differs between two years or two scenarios, split it into factors, exactly like the Kaya identity in Go 1 but for transport. For each mode *m*:

$$CO_2 = \Sigma_m \, A \cdot S_m \cdot I_m \cdot F_m \tag{3.8}$$

with A total activity (pkm), S_m the mode share, I_m energy per pkm and F_m CO_2 per unit of energy. The **logarithmic mean Divisia index** (LMDI) attributes the change to each factor with no leftover residual:

$$\Delta CO_2(X) = \Sigma_m \, L(C_m^{\,1}, C_m^{\,0}) \cdot \ln(X_m^{\,1} \,/\, X_m^{\,0}), \quad L(a, b) = (a - b) \,/\, (\ln a - \ln b) \tag{3.9}$$

SIMPLE CALCULATION

LMDI for an illustrative country (three modes, two years)

1. CO_2 falls from 19.36 to 13.13 Mt: a change of -6.24 Mt.
2. Activity (pkm 200 → 300 bn): +6.49 Mt.
3. Mode structure (cars 80 % → 70 %, rail 5 % → 15 %): -1.72 Mt.
4. Energy intensity (cars 1.6 → 1.1 MJ/pkm, etc.): -5.71 Mt.
5. Carbon intensity (fuel switching, cleaner electricity): -5.30 Mt.
6. Sum of effects: -6.24 Mt — equal to the total change, with no residual.

Answer: intensity and carbon each cut more than mode shift; activity growth eats most of the gains

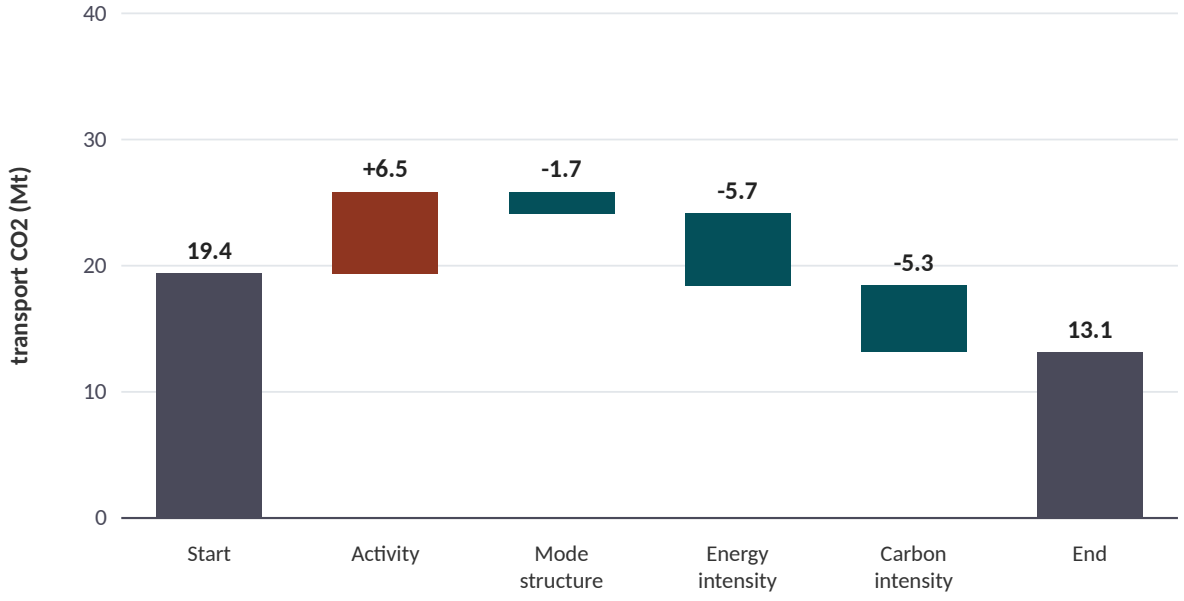


Figure 3.9 — LMDI waterfall for the example. What to notice: the bars sum exactly to the total change. This is the standard way IAM transport studies (and module 3.3's comparisons) report what drove mitigation.

BUILD IT · LMDI ON GCAM OUTPUT

```
def L(a, b):
    a, b = max(a, EPS), max(b, EPS)
    return a if abs(a - b) < 1e-15 else (a - b) / (math.log(a) - math.log(b))


def factors(svc, en, scen, year):
    s = svc[(svc.scenario == scen) & (svc.year == year)].groupby("mode").value.sum()
    e = en[(en.scenario == scen) & (en.year == year)].copy()
    e["co2"] = e.apply(lambda r: r.value * 1e3 * EF.get(r.fuel, 0.0), axis=1)     # EJ -> Mt
CO2
    ej = e.groupby("mode").value.sum(); co2 = e.groupby("mode").co2.sum()
    A = s.sum(); out = {}
    for m in ej.index:
        out[m] = dict(A=A, S=s[m] / A, I=ej[m] / s[m], C=(co2[m] / ej[m]) if ej[m] > 0 else
0.0, CO2=co2[m])
    return out


def lmdi(f0, f1):
    eff = dict(activity=0.0, structure=0.0, intensity=0.0, carbon=0.0)
    for m in set(f0) | set(f1):
        a = f0.get(m, dict(A=1, S=EPS, I=EPS, C=EPS, CO2=0.0)); b = f1.get(m, dict(A=1,
S=EPS, I=EPS, C=EPS, CO2=0.0))
        if a["CO2"] <= 0 and b["CO2"] <= 0:
            continue
        w = L(b["CO2"], a["CO2"])
        for k, x in (("activity", "A"), ("structure", "S"), ("intensity", "I"), ("carbon",
"C")):
            eff[k] += w * math.log(max(b[x], EPS) / max(a[x], EPS))
    tot0 = sum(v["CO2"] for v in f0.values()); tot1 = sum(v["CO2"] for v in f1.values())
    eff.update(co2_start=tot0, co2_end=tot1, change=tot1 - tot0, residual=tot1 - tot0 -
sum(eff[k] for k in ("activity", "structure", "intensity", "carbon")))
    return eff
…
```

Step 4: stress-test

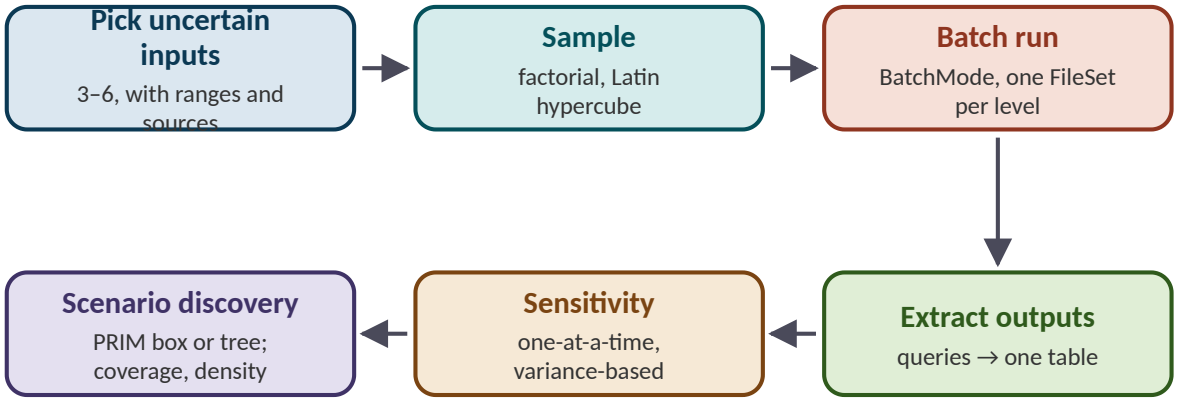


Figure 3.10 — From uncertain inputs to robust findings. What to notice: the analysis methods only work on a table built the same way for every run (step 1).

One at a time: the tornado

Move each uncertain input from its low to its high value with everything else central, and plot the range of the output. The longest bar is the input your conclusion depends on most. It is cheap (two runs per input) but misses interactions.

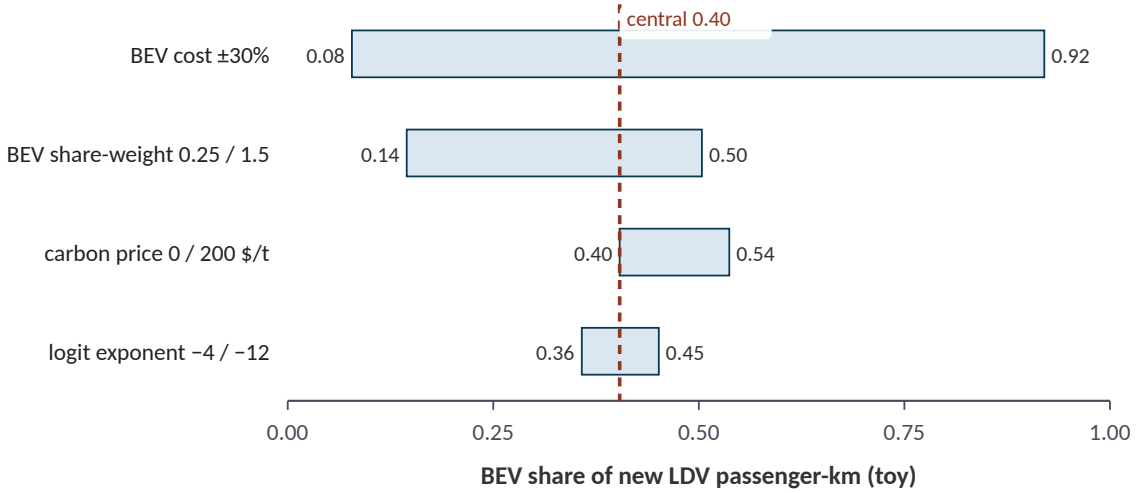


Figure 3.11 — Toy tornado for the BEV share of new car pkm in 2050 (central 0.40). What to notice: BEV cost dominates, and the share-weight path — an assumption, not data — is second. A \$200/tCO_2 carbon price moves the result less, because fuel is a small part of the cost per pkm (Figure 3.3); the logit exponent matters least here because the two costs are close.

Input (low / high)	BEV share at low	at high	range
BEV cost ±30%	0.08	0.92	0.84
BEV share-weight 0.25 / 1.5	0.14	0.50	0.36
carbon price 0 / 200 $/t	0.40	0.54	0.13
logit exponent −4 / −12	0.45	0.36	0.09

Many at once: ensembles and scenario discovery

Run a full factorial or Latin-hypercube ensemble in batch mode. Label each run "high" or "not high" for the outcome you care about (for example, 2050 transport CO_2 above a target). Then search for a simple box in input space that captures the high runs — with the PRIM algorithm or a classification tree (lab 13 uses `rpart`). Judge the box with two numbers:

$$\text{coverage} = (\text{high runs inside the box}) \,/\, (\text{all high runs}), \quad \text{density} = (\text{high runs inside the box}) \,/\, (\text{all runs inside the box}) \tag{3.10}$$

SIMPLE CALCULATION

Scenario discovery on 36 runs

1. 9 of the 36 runs miss the 2050 target ("high").
2. Candidate box: "BEV cost high AND share-weight path slow" contains 10 runs, of which 7 are high.
3. Coverage = 7 ÷ 9 = 0.78: the box explains 78 % of the failures.
4. Density = 7 ÷ 10 = 0.70: inside the box, 70 % of runs fail.
5. A good box has both high; narrowing the box raises density but lowers coverage — report the trade-off.

Answer: coverage 0.78, density 0.70

BUILD IT · SCENARIO DISCOVERY

```
tree = DecisionTreeClassifier(max_depth=3, min_samples_leaf=10,
random_state=0).fit(df[X_COLS], df.high)
lines += ["Classification tree (max_depth 3):", export_text(tree, feature_names=X_COLS,
decimals=3)]
imp = sorted(zip(X_COLS, tree.feature_importances_), key=lambda t: -t[1])
lines += ["Feature importance: " + ", ".join(f"{k} {v:.2f}" for k, v in imp), ""]

# PRIM-style peeling
box = {c: [df[c].min(), df[c].max()] for c in X_COLS}
inside = np.ones(len(df), bool); alpha = 0.05; path = []
def stats(mask):
    h = df.high[mask].sum(); return h / df.high.sum(), (h / mask.sum() if mask.sum() else 0)
while True:
    cov, den = stats(inside); path.append((cov, den, {k: tuple(v) for k, v in box.items()}))
    best = None
    for c in X_COLS:
…
```

WHAT IT PRINTS (300-RUN SYNTHETIC ENSEMBLE)

```
300 runs; 'high' = 2050 CO2 above 46.3 Mt (top quarter): 75 runs

Classification tree (max_depth 3):
|--- bev_cost_mult <= 0.910
|   |--- bev_cost_mult <= 0.883
|   |   |--- carbon_start_usd_tco2 <= 9.315
|   |   |   |--- class: False
|   |   |--- carbon_start_usd_tco2 >  9.315
|   |   |   |--- class: False
|   |--- bev_cost_mult >  0.883
|   |   |--- class: False
|--- bev_cost_mult >  0.910
|   |--- carbon_start_usd_tco2 <= 130.935
|   |   |--- bev_cost_mult <= 0.993
|   |   |   |--- class: True
|   |   |--- bev_cost_mult >  0.993
|   |   |   |--- class: True
|   |--- carbon_start_usd_tco2 >  130.935
|   |   |--- class: False

Feature importance: bev_cost_mult 0.87, carbon_start_usd_tco2 0.13, car_lf_2050 0.00,
gdp_growth 0.00

PRIM-style box after 28 peels: coverage 0.80, density 0.92 (base rate 0.25)
Restricted inputs:
  bev_cost_mult: 0.940 to 1.094  (full range 0.503 to 1.094)
  carbon_start_usd_tco2: 0.230 to 124.658  (full range 0.230 to 149.950)
```

Read the output like this: the tree splits first on BEV cost, and the PRIM-style box keeps 80 % of the high runs with 92 % of runs inside it high (against a base rate of 25 %). The car load factor and GDP growth were never used — in this synthetic ensemble, they do not explain failure. With real GCAM runs, check that the box is stable when you re-sample.

Step 5: benchmark against other models

Put your Reference and policy runs next to the AR6 scenario database (or the newer CMIP7 ScenarioMIP set) for the same region and variables: `Final Energy|Transportation`, `Emissions|CO2|Energy|Demand|Transportation`, `Energy Service|Transportation|Passenger`. Lab 12 shows how to filter AR6 runs to a similar carbon price. Being outside the range is not wrong, but you must be able to say why (module 3.3).

TRAP

Reporting a percentage of the wrong base. "Transport CO_2 falls 60 %" — below what? The 2020 level, or the 2050 Reference? The two can differ by a factor of two when Reference emissions grow. Always say "relative to Reference in the same year" or "relative to 2020".

TRY IT NOW

1. Using Eq. 3.9, what sign does the activity effect have if pkm stays constant? What if pkm falls?
2. Your tree's best box has coverage 0.9 and density 0.3. Is it a useful finding? What would you try next?
3. In Figure 3.11, which bar would you test first in a full GCAM ensemble, and why?

RECAP SO FAR

1. **3.1 GCAM as research software:** data → gcamdata → XML → C++ core → database → queries; add-ons load in order; check the log before you trust a number.
2. **3.2 Transport in GCAM:** four final demands; cost per pkm = (fuel × intensity + non-fuel) ÷ load factor, plus wage × multiplier ÷ speed at mode level; modes share by relative-cost logit, technologies by absolute-cost logit.
3. **3.3 Literature:** IAM transport mitigation comes mostly from intensity and fuel switching; demand and behaviour are under-represented; GCAM-KSA puts Saudi transport at 115 Mt in 2060 under net zero.
4. **3.4 Study design:** one question, one decision metric; Reference + levers × policies; plan uncertainty before running.
5. **3.5 Building scenarios:** add-on XML for costs, load factors, share-weights (with interpolation rules) and policies; mind the v8.5 units.
6. **3.6 Analysis:** report differences from Reference; LMDI splits CO2 into activity, structure, intensity, carbon; tornado and scenario discovery show what matters.

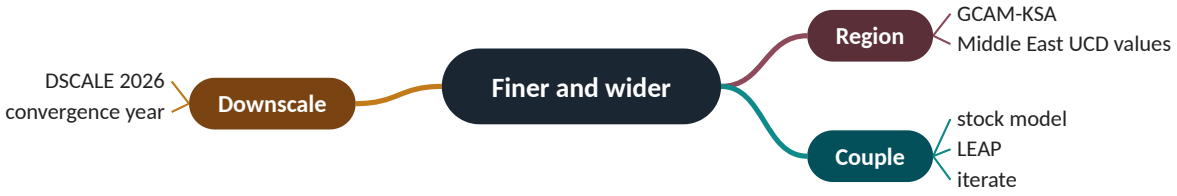
MODULE 7 OF 8

Going finer and wider

Core GCAM sees Saudi Arabia as part of "Middle East", and sees cars as a flow of new purchases. Research often needs a country, a city, or a fleet.

HOOK

In GCAM-KSA, the version of GCAM in which KAPSARC and PNNL split Saudi Arabia out of the Middle East, Saudi transport CO_2 in 2060 ranges from 418 Mt with no policy to 115 Mt under net zero — against 176 Mt in 2015. Even in the net-zero case, around 80 % of new cars are zero- or low-emission, and aviation and shipping dominate what is left. None of these numbers can be read from core GCAM, where Saudi Arabia shares its transport parameters with Iran, Iraq and the rest of the region.



A country of its own: the GCAM-KSA approach

Core GCAM has 32 regions; Saudi Arabia is inside "Middle East", and its transport parameters come from the UC Davis "Middle East" region (`energy/mappings/UCD_ctry.csv`). GCAM-KSA v1.0 is GCAM 6.0 with Saudi Arabia as a 33rd region, calibrated to 2015 with KAPSARC data. To build a country region yourself:

1. Add the country to `common/iso_GCAM_regID.csv` and the region-name table, so gcamdata splits it from its parent region.
2. Check every mapping that is by country or by UCD region (transport uses `UCD_ctry.csv`) so the new region gets parameters.
3. Override region-average values with national data where you have it: fleet, efficiency, load factors, fuel prices, elasticities (KAPSARC used its transport and infrastructure team's data and Saudi literature such as fuel-price elasticity studies).
4. Re-check calibration: base-year transport energy by fuel must match national balances.

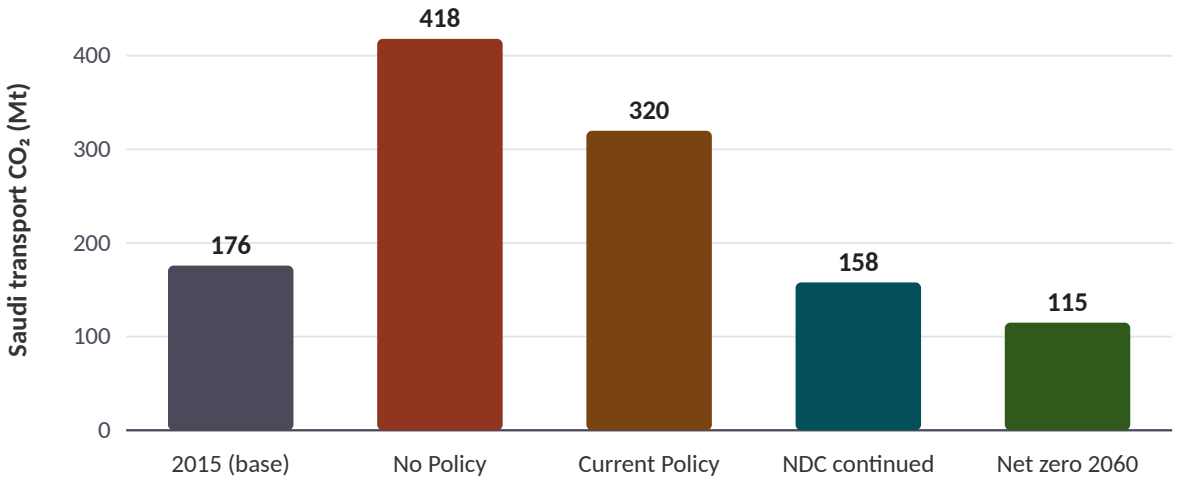


Figure 3.12 — Saudi transport CO_2 in 2060 by scenario in GCAM-KSA (Kamboj et al. 2024, KAPSARC discussion paper KS--2024-DP01). What to notice: the gap between No Policy and Current Policy (98 Mt) is smaller than the gap between Current Policy and NDC continued (162 Mt); most of the net-zero reduction is already delivered by continuing the NDC path.

GCAM-KSA study (verified)	Transport-relevant finding
Kamboj et al. (2024) *Energy Strategy Reviews* 55, 101537, doi:10.1016/j.esr.2024.101537	NDC must roughly triple for net zero 2060; transport and industry are the hardest sectors; carbon removal needed
Kamboj et al. (2024) KAPSARC DP, doi:10.30573/KS--2024-DP01 (DOI resolution not confirmed)	2060 transport CO_2 418 / 320 / 158 / 115 Mt; ≈80 % of new cars and ≈65 % of new trucks ZEV or low-emission in net zero
Apeaning et al. (2025) *Energy and Climate Change* 6, 100184, doi:10.1016/j.egycc.2025.100184	Early action cuts the policy cost of net zero by 38–72 % versus delay
Qiu et al. (2024) *Environ. Res. Lett.* 19, 064042 (DOI not confirmed)	Direct air capture removes 0.28–0.33 Gt/yr by 2060 — the backstop for aviation and shipping residuals
KAPSARC (2025) *Decarbonizing Road Freight Transportation in Saudi Arabia* (grey)	Land freight CO_2 29.4 Mt in 2060 under net zero; rail 30.6 % and ZEVs 25.9 % of land freight

RESEARCH NOTE

Access. The GCAM-KSA documentation (KAPSARC, dated 2 June 2026) sits behind a registration form and was not read for this handbook; GCAM-KSA itself is not part of the public gcam-core release. Students without access can (a) work with the Middle East region of core GCAM and state that Saudi values are regional averages, or (b) build a country split as above, which is a substantial project in itself.

Downscaling: from a region to its countries

When building a new region is too expensive, results can be **downscaled**: a regional pathway is split among countries using country data, with the rule that the countries add back to the region. The simplest rule keeps today's shares; better rules let shares move.

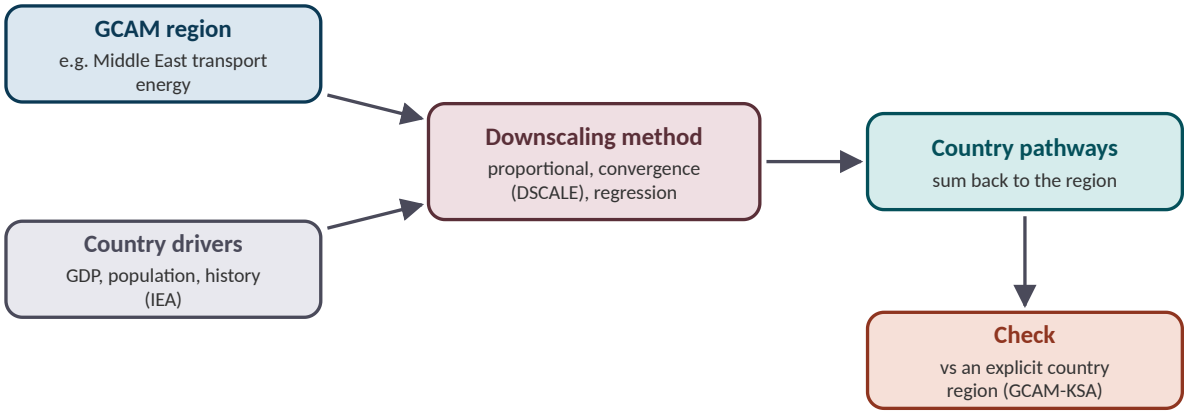


Figure 3.13 — A downscaling workflow. What to notice: the check against an explicit country region (if you have one, like GCAM-KSA) is the only true test of a downscaling method.

DSCALE (Sferra et al. 2026, *Geosci. Model Dev.* 19, 3157–3191, doi:10.5194/gmd-19-3157-2026) downscales regional mitigation pathways to 143 countries. Its central idea is **convergence**: each country starts on its own trend and converges towards the regional pattern by a chosen year. In its simplest form, for a country's share s of a regional variable:

$$s_c(t) = (1 - w_t) \cdot s_c(t_0) + w_t \cdot s_c^*, \quad w_t = \min(1, (t - t_0) \,/\, (T_{conv} - t_0)) \tag{3.11}$$

This linear blend is a teaching simplification of DSCALE's method, which blends a national-trend projection with the regional IAM pathway, reconciles energy variables by iterated proportional rescaling and treats emissions separately. Read the paper before using the method in research.

SIMPLE CALCULATION

Downscale Saudi Arabia's share of Middle East transport energy

1. Starting share in 2020: 0.50 (illustrative). Long-run target share s^* (e.g. from GDP share): 0.42.
2. Convergence by 2060: in 2050, w = (2050 − 2020) ÷ (2060 − 2020) = 0.75, so s = 0.25 × 0.50 + 0.75 × 0.42 = 0.44.
3. Convergence by 2100: w = 30 ÷ 80 = 0.375, so s = 0.625 × 0.50 + 0.375 × 0.42 = 0.47.
4. Multiply by the regional total; repeat for the other countries and rescale so shares sum to 1.

Answer: 0.44 or 0.47 of the region in 2050 — the convergence year is itself an uncertain assumption

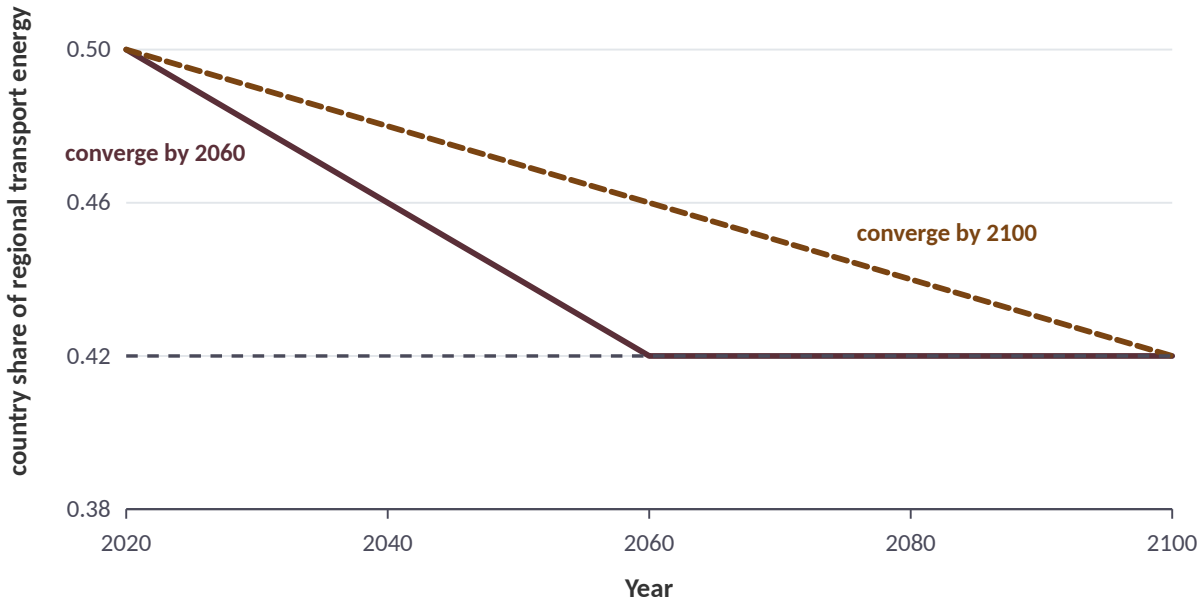


Figure 3.14 — The same country downscaled with two convergence years. What to notice: the choice is invisible in the regional total but moves the country result for decades; test it as a sensitivity (research question RQ7).

BUILD IT · DOWNSCALING DEMO

```
    gdp = C.gdp0 * (1 + C.growth) ** t
    # method 1: proportional
    prop = REG[y] * C.energy0 / C.energy0.sum()
    # method 2: convergence of energy intensity toward the regional intensity, then
harmonise
    ei0 = C.energy0 / C.gdp0
    reg_ei = REG[y] / gdp.sum()
    w = min(1.0, t / (CONV_YEAR - 2020))
    ei = np.exp((1 - w) * np.log(ei0) + w * math.log(reg_ei))
    raw = ei * gdp
    conv = raw * REG[y] / raw.sum()                    # harmonised to the GCAM regional
total
    for i, c in enumerate(C.country):
        rows.append(dict(year=y, country=c, gdp=gdp[i], proportional_EJ=prop[i],
convergence_EJ=conv[i],
                         convergence_unharmonised_EJ=raw[i]))
df = pd.DataFrame(rows)
df.to_csv(os.path.join(HERE, "downscale_results.csv"), index=False)

chk = df.groupby("year")[["proportional_EJ", "convergence_EJ"]].sum().sub(REG,
axis=0).abs().max().max()
print(f"Regional totals reproduced by both methods: max error {chk:.1e} EJ")
show = df[df.year.isin([2020, 2050, 2100])].pivot_table(index="country", columns="year",
                                                        values=["proportional_EJ",
"convergence_EJ"]).round(3)
print(show.to_string())
harm = df[df.year == 2050].convergence_EJ.sum() / df[df.year ==
2050].convergence_unharmonised_EJ.sum()
…
```

The kit's demo goes one step beyond Eq. 3.11: it converges each country's **energy intensity** (energy per unit of GDP) log-linearly towards the regional intensity, multiplies by each country's own GDP path, and then **harmonises** — rescales — so the countries add up exactly to the GCAM regional total. On its synthetic six-

country region both methods reproduce the regional total to within 10^{-15} EJ, but they disagree strongly by 2100: the fast-growing Country B gets 0.98 EJ under convergence against 0.75 EJ under fixed shares. The 2050 harmonisation factor of 0.60 shows how far the unharmonised country paths overshoot the region — a number worth reporting.

Fleets: what GCAM does not count

GCAM tracks transport as **service output** by technology and vintage of service, not as vehicles. The query "transport service output by tech (new)" is the nearest thing to sales, but there are no vehicle counts, survival curves or second-hand markets. The difference between new sales and the vehicles on the road is large and slow:

SIMPLE CALCULATION

Sales versus stock with a 15-year vehicle life

1. Battery-car share of new sales follows an s-curve passing 50 % in 2035.
2. Stock share in 2035 = average of the last 15 years of sales shares = 0.15.
3. In 2050 sales are 99.5 % battery cars, but the stock is only 88 %.
4. Fuel use and CO_2 follow the stock, not the sales.

Answer: stock lags sales by roughly a decade: 15 % of cars on the road when half of sales are electric

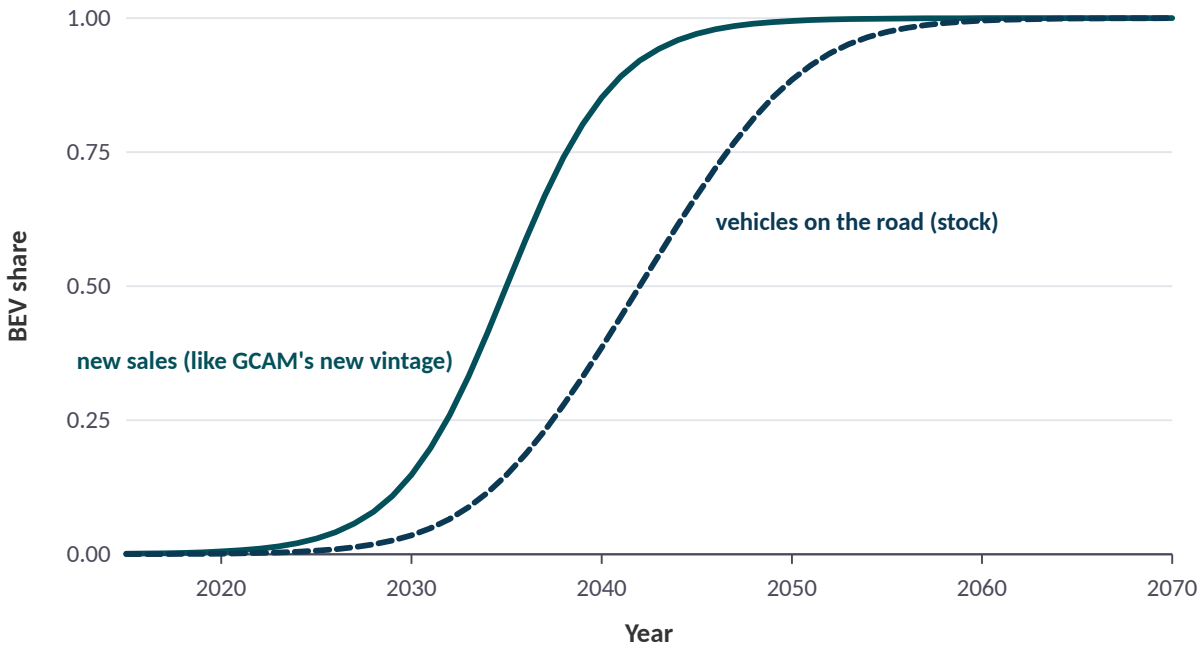


Figure 3.15 — New sales versus the vehicles on the road (rectangular 15-year lifetime, illustrative). What to notice: a policy that changes sales today changes emissions mostly in the 2040s. GCAM's vintaging captures part of this lag, but not with vehicle counts.

Coupling GCAM with other models

When a question needs detail GCAM does not have — fleet counts, hourly charging load, national policy detail, behaviour by consumer group — the usual answer is a **soft link**: two models exchange numbers through files, run in turn, until they agree.

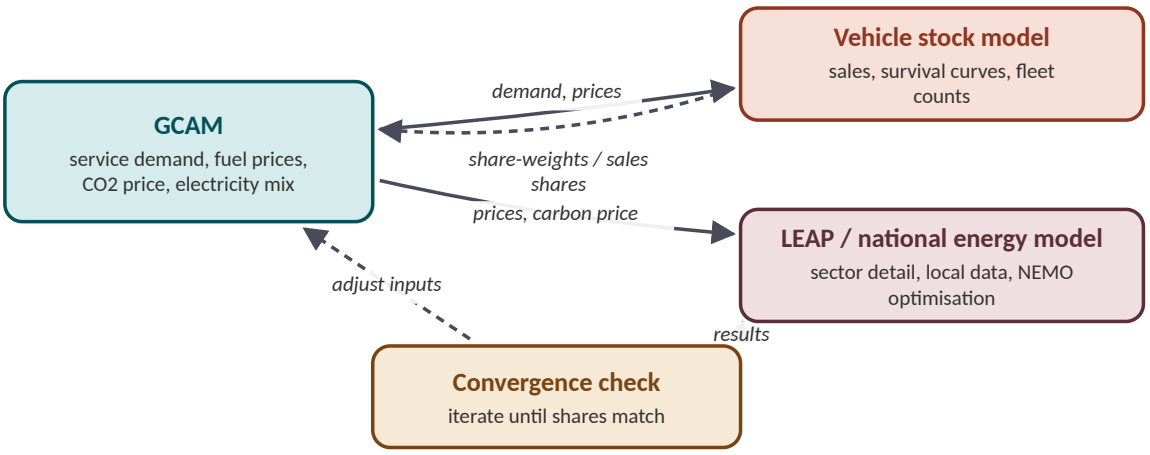


Figure 3.16 — Soft-linking GCAM with a vehicle-stock model and a national model such as LEAP. What to notice: the feedback arrows (dashed) are what make it coupling rather than a one-way handover; without them, the two models describe different worlds.

Partner model	What it adds	What passes from GCAM	What passes back
Vehicle stock-turnover model	fleet counts, sales, survival, stock lag	service demand by mode, fuel prices	sales shares → share-weights or vintage adjustments
LEAP (with NEMO optimisation)	national sector detail, local data, power dispatch	fuel and carbon prices, regional demand trends	national electricity mix, demand corrections
MESSAGEix-Transport	consumer groups with disutility costs (McCollum et al. 2017)	—(comparison)	behavioural parameters to emulate in GCAM (RQ10)
Hourly power / charging model	peak load, flexibility	transport electricity (`elect_td_trn`)	storage and grid cost signals

RESEARCH NOTE

Convergence is the hard part. Decide in advance which variables must agree (for example, new-vintage BEV share within 2 percentage points in every period), the maximum number of iterations, and what you report if they do not converge. Coupled-model papers that skip this step usually show one model's result labelled as both.

TRAP

Downscaled numbers are not observations. A country pathway from a downscaling method inherits every assumption of the regional run plus those of the method. Never compare it with a national model as if one were "data"; compare both with history and with each other, and report the convergence year.

TRY IT NOW

1. Recompute the 2050 Saudi share with convergence by 2040. Why does it no longer change after 2040?
2. With a 12-year vehicle life instead of 15, is the 2035 stock share higher or lower than 0.15? Explain without calculating.
3. Sketch the files that would pass between GCAM and LEAP-KSA in one iteration, naming one query and one input element.

RECAP SO FAR

1. **3.1 GCAM as research software:** data → gcamdata → XML → C++ core → database → queries; add-ons load in order; check the log before you trust a number.
2. **3.2 Transport in GCAM:** four final demands; cost per pkm = (fuel × intensity + non-fuel) ÷ load factor, plus wage × multiplier ÷ speed at mode level; modes share by relative-cost logit, technologies by absolute-cost logit.
3. **3.3 Literature:** IAM transport mitigation comes mostly from intensity and fuel switching; demand and behaviour are under-represented; GCAM-KSA puts Saudi transport at 115 Mt in 2060 under net zero.
4. **3.4 Study design:** one question, one decision metric; Reference + levers × policies; plan uncertainty before running.
5. **3.5 Building scenarios:** add-on XML for costs, load factors, share-weights (with interpolation rules) and policies; mind the v8.5 units.
6. **3.6 Analysis:** report differences from Reference; LMDI splits CO2 into activity, structure, intensity, carbon; tornado and scenario discovery show what matters.
7. **3.7 Finer and wider:** GCAM-KSA splits Saudi Arabia from the Middle East; DSCALE downscales by convergence; soft-links add vehicle stocks or national detail.

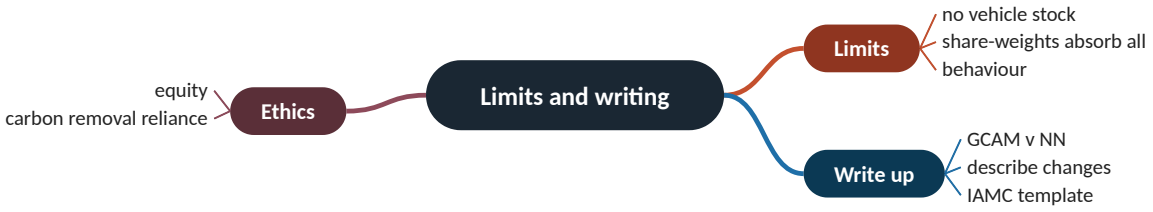
MODULE 8 OF 8

Limits, ethics and writing it up

A transport study with GCAM is credible when it says plainly what the model can and cannot see, and when someone else can rebuild every number.

HOOK

A net-zero transport pathway with 80 % zero-emission car sales can still leave aviation, shipping and heavy freight emitting tens of megatonnes a year, balanced by direct air capture that does not yet exist at scale. The model will report it as "net zero". Whether that is a plan or a gamble is not a question the model answers — and your paper has to say so.



What GCAM transport cannot see

Limit	Why it matters	What you can do
No vehicle stock or counts	sales and fleet lag (Figure 3.15) are not explicit; "sales" is a proxy	report "service output by tech (new)" as a sales proxy; couple a stock model (RQ9)
Share-weights absorb what the costs miss	infrastructure, preferences, regulation and data error sit in one number	report share-weight paths as assumptions; test them (Figure 3.11)
Behaviour is a single representative traveller	no income groups, early adopters or range anxiety	add disutility costs by consumer group (RQ10); compare with MESSAGEix-Transport
Freight has no value of time	freight mode shift responds to money cost only	test the freight logit exponent; add rail speed explicitly
Aviation and shipping use generic liquids	a SAF or e-fuel mandate cannot be targeted without model changes	add a dedicated pass-through sector (RQ5); state the simplification
Regional parameters from 14 UCD regions	Saudi values are Middle East averages in core GCAM	national overrides or a country region (module 3.7)
Demand from income and price only	urban form, telework, infrastructure and policy-driven demand reduction are outside the model	exogenous demand scenarios; low-demand sensitivities (Grubler et al. 2018 style)

Ethics: whose future, whose cost?

- **Reliance on carbon removal.** Net-zero transport in GCAM-KSA leans on direct air capture for aviation and shipping residuals (Qiu et al. 2024). Report gross and net emissions separately, and show a case with limited removal.

- **Distribution.** A carbon price or fuel-price reform raises the cost of travel for households that cannot switch. GCAM's single representative consumer cannot show who pays; say so and point to studies that can.
- **Oil exporters.** For Saudi Arabia, transport decarbonisation interacts with domestic fuel-price reform and export revenues; a cost-minimising pathway is not automatically a socially acceptable one.
- **Authority of numbers.** A single scenario figure is easily quoted as a forecast. Show ranges, label scenarios by assumptions, and include "what the model cannot tell us" in the abstract, not only in the appendix.

Describing GCAM in a paper

JGCRI publishes suggested community guidelines for peer-reviewed articles that use GCAM (community-guide.html, checked 23 October 2026). In short:

Guideline	In your paper
State the version	"We use GCAM v9.1 (tag gcam-v9.1)" — never just "GCAM"
Describe the model in a short paragraph	multisector model developed at JGCRI, open source; 32 energy-economy regions, land in more than 300 subregions (396 in the guideline text), 235 water basins; cite JGCRI (2025), GCAM documentation v8.2, doi:10.5281/zenodo.15581183, plus model papers relevant to your sector
Data and code availability	cite the public repository of the version used, with its DOI
Describe every modification	list every departure from the public release; for substantial changes, rename the model with an identifier, e.g. "GCAM-<institution>" (as GCAM-KSA does)
Ask and collaborate	questions go to GCAM GitHub Discussions; JGCRI team members may be invited as co-authors

RESEARCH NOTE

Two numbers in the guideline. The community guideline says 396 land subregions, while the documentation overview says "more than 300" (the counts differ between pages and versions). Quote the number that matches your version's input files and say where it comes from.

Reporting in the IAMC format

Report results in the IAMC template (model, scenario, region, variable, unit, then one column per year) with standard variable names, so they can be compared with the AR6 and CMIP7 scenario databases and read with **pyam**. Typical transport variables:

IAMC variable	GCAM source
Energy Service|Transportation|Passenger (bn pkm/yr)	transport service output by service (trn_pass)
Energy Service|Transportation|Freight (bn tkm/yr)	transport service output by service (trn_freight)
Final Energy|Transportation|Electricity / Liquids / Hydrogen / Gases (EJ/yr)	transport final energy by fuel
Emissions|CO2|Energy|Demand|Transportation (Mt CO2/yr)	CO2 emissions by sector (excluding resource production)
Price|Carbon (US$2010/t CO2)	CO2 prices (converted from 1990$/tC)

BUILD IT · REPORTING CHECKLIST

```
# Reporting checklist for a GCAM transport study

Tick every line before submitting a report, thesis chapter or paper.

## A. Model description (GCAM community guidelines, https://jgcri.github.io/gcam-
doc/community-guide.html)
- [ ] GCAM version stated ("GCAM v9.1"), with the public repository and the DOI of that
release cited.
- [ ] Documentation cited (the guidelines suggest "JGCRI, 2025. GCAM Documentation (Version
8.2)" with its Zenodo DOI; note that the docs describe v8.2 while the release used is 9.x).
- [ ] Standard introductory description of GCAM used, with links to the documentation.
- [ ] Every change from the public model listed (file, element, old and new value, reason).
Substantial changes: model renamed "GCAM-<institution>" and the change stated in the opening
paragraph.
- [ ] GCAM team members invited as co-authors where their expertise is used (guidelines);
questions on data or results raised on GCAM GitHub Discussions.

## B. Transport-specific transparency
- [ ] Nesting and logit exponents used (default from A54.sector.csv and
A54.tranSubsector_logit_revised.csv, or changed).
- [ ] Source of vehicle costs and intensities named (OTAQ_trn_data_EMF37.csv overrides
UCD_trn_data_CORE.csv for most modes in 9.1).
- [ ] Units stated: pkm/tkm in billions, costs 1975$/vkm, intensities EJ per billion vkm,
carbon prices 1990$/tC (converted for the reader, e.g. to 2024$/tCO2, with the deflator
named).
- [ ] Share-weight changes reported as assumptions, with the reason.
- [ ] Statement that GCAM has no vehicle stock: "sales shares" are new-vintage output
shares.
- [ ] Value-of-time settings (time-value multipliers, speeds, wage definition) if mode shift
matters to the result.

## C. Results reporting (IAMC-style variable names)
Report regional results in the IAMC template format (model, scenario, region, variable,
unit, years) so they can be compared with other models. Names below follow the IAMC/AR6
template convention; check them against the current IAMC common definitions
(https://github.com/IAMconsortium/common-definitions) before submission.
| Variable | Unit | GCAM source query |
| --- | --- | --- |
| Final Energy\|Transportation | EJ/yr | transport final energy by fuel (sum) |
| Final Energy\|Transportation\|Liquids | EJ/yr | ... fuel = refined liquids enduse |
| Final Energy\|Transportation\|Electricity | EJ/yr | ... fuel = elect_td_trn |
| Final Energy\|Transportation\|Hydrogen | EJ/yr | ... fuel = H2 LDV, H2 MHDV, LH2 |
| Final Energy\|Transportation\|Gases | EJ/yr | ... fuel = delivered gas |
| Energy Service\|Transportation\|Passenger | bn pkm/yr | transport service output by
service |
| Energy Service\|Transportation\|Freight | bn tkm/yr | transport service output by service
|
…
```

The reproducible package

Publish (or archive with a DOI, e.g. on Zenodo) a package that lets a reader rebuild every figure: the GCAM version and commit; every add-on XML and configuration or batch file; any gcamdata modification chunk; the query file; the analysis scripts; the extracted results table; and a README with the run log (module 3.1). Large output databases do not need to be shared if the extracted table and the means to rebuild it are.

TRAP

"Results available on request." For a model that anyone can download, this is no longer acceptable to many journals or reviewers. If a reader cannot rebuild your scenarios from the paper and its archive, the scenarios are not reproducible — however carefully they were run.

TRY IT NOW

1. Write the two-sentence model description for a paper using GCAM v9.1 with a modified Middle East transport sector. What name would you give the modified model?
2. List the five files a reviewer would need to rebuild your carbon-price scenario.
3. Pick one limit from the first table that matters most for a Saudi ZEV-mandate study, and write the sentence you would put in the abstract about it.

RECAP SO FAR

1. **3.1 GCAM as research software:** data → gcamdata → XML → C++ core → database → queries; add-ons load in order; check the log before you trust a number.
2. **3.2 Transport in GCAM:** four final demands; cost per pkm = (fuel × intensity + non-fuel) ÷ load factor, plus wage × multiplier ÷ speed at mode level; modes share by relative-cost logit, technologies by absolute-cost logit.
3. **3.3 Literature:** IAM transport mitigation comes mostly from intensity and fuel switching; demand and behaviour are under-represented; GCAM-KSA puts Saudi transport at 115 Mt in 2060 under net zero.
4. **3.4 Study design:** one question, one decision metric; Reference + levers × policies; plan uncertainty before running.
5. **3.5 Building scenarios:** add-on XML for costs, load factors, share-weights (with interpolation rules) and policies; mind the v8.5 units.
6. **3.6 Analysis:** report differences from Reference; LMDI splits CO2 into activity, structure, intensity, carbon; tornado and scenario discovery show what matters.
7. **3.7 Finer and wider:** GCAM-KSA splits Saudi Arabia from the Middle East; DSCALE downscales by convergence; soft-links add vehicle stocks or national detail.
8. **3.8 Limits and writing:** say "GCAM v NN", describe every departure, publish code and data, report in the IAMC format.

Master mind map

The whole go on one page: one card per module. Cover a card and rebuild its four points from memory before your project starts.

3.1 GCAM as research software

- gcamdata → input XML → C++ core → BaseX database → queries
- markets cleared in log prices by Broyden's method; recursive-dynamic
- check: solver log, calibration, the changed variable
- record version, commit, configuration and a run log

3.2 Transport in GCAM

- four final demands; nests down to drivetrains
- $P = (P_f \cdot I + N) \div L$, plus wage × multiplier ÷ speed
- modes: relative-cost logit; drivetrains: absolute-cost (−8 for cars)
- demand: income elasticity 1.0, price −1.25 (trn_pass)

3.3 What the literature knows

- intensity and fuel switching do most of the work
- demand and behaviour under-represented
- GCAM-KSA: 115 Mt transport CO_2 in 2060 under net zero
- always cite the model version

3.4 Designing a study

- one decision, one metric, one comparison
- background world + levers + policies
- uncertainty table with sources; count the runs
- hindcast, benchmark, face validity

3.5 Building scenarios

- add-ons after the core file; exact names
- v8.5 units: 1975$, EJ per bn vkm, bn pkm
- share-weights need interpolation rules
- fixedTax, constraint, credit-market standards; batch files

3.6 Analysis and stress tests

- one script, one tidy table, all scenarios
- differences from Reference; LMDI with no residual
- tornado: BEV cost, then share-weight path
- scenario discovery: coverage and density

3.7 Finer and wider

- country region (GCAM-KSA) or downscaling
- convergence year is an assumption
- stock lags sales by about a decade
- soft-link with stock models or LEAP; agree on convergence

3.8 Limits, ethics, writing

- no fleet, one traveller, generic aviation liquids
- gross vs net; who pays
- "GCAM v NN", every change, repository DOI
- IAMC format; a package that rebuilds every figure

Ten research questions

Each question below is answerable with GCAM and the kit in a term-length project, and each touches a gap named in module 3.3. Pick one, sharpen it with module 3.4, and pre-register your design.

RQ1. How sensitive are global and Saudi BEV pathways to share-weight versus cost assumptions?

Method	Data	Outputs
Factorial on BEV non-fuel cost (±30 %) × share-weight path (v5.4-style vs current s-curve) × carbon price; variance (Sobol) decomposition	BNEF pack prices; IEA Global EV Outlook (CC BY) for 2020–2025 checks	BEV share of new LDV pkm ("service output by tech (new)"); variance shares

RQ2. Can a Saudi-resolved GCAM reproduce the 2015–2024 gasoline demand response to the 2016/2018 price reforms?

Method	Data	Outputs
Hindcast; fit price and income elasticities of trn_pass to observed demand	KAPSARC data portal, GASTAT, published Saudi elasticity studies	fitted elasticities; 2060 LDV demand and CO_2 under fitted vs default values

RQ3. A Saudi ZEV mandate versus a carbon price: which is cheaper for the same land-transport CO_2 cut?

Method	Data	Outputs
Credit-based ZEV standard (policy-portfolio-standard + res-secondary-output) vs equal-CO_2 carbon price	Vision 2030 and SEEC targets for scenario design	credit price, transport CO_2, electricity and H_2 demand, abatement cost

RQ4. What does freight rail expansion buy in Saudi Arabia?

Method	Data	Outputs
Raise Freight Rail share-weight and electric-rail availability; test the freight logit exponent (−2 → −4); add a heavy-duty efficiency standard	KAPSARC freight study (2025)	tkm by mode, freight CO_2; comparison with rail 30.6 % in KAPSARC's net zero

RQ5. Aviation and shipping residuals for a Gulf hub economy: bio-jet, e-fuels or direct air capture?

Method	Data	Outputs
Aviation-specific liquids pass-through with a SAF credit mandate; add an e-fuel technology (Speizer et al. 2024 approach); compare with DACCS offsets	ICAO/IATA traffic; IMO Fourth GHG Study; Speizer et al. 2024 supplement	break-even CO_2 prices, residual emissions, land and water footprint

RQ6. Does value-of-time realism (speed growth, metro speed) change mode shares in Riyadh-type cities?

Method	Data	Outputs
Scenarios on rail and bus speed and time-value multipliers	Riyadh public transport ridership; travel surveys	pkm by mode, LDV energy, sensitivity to the value of time

RQ7. How consistent are downscaled GCC country transport pathways with an explicit country region?

Method	Data	Outputs
Downscale Middle East transport energy (DSCALE-style); compare downscaled Saudi results with a Saudi-resolved run; vary the convergence year	GCAM outputs; IEA country balances; SSP GDP and population	downscaling error metrics; UAE, Kuwait, Qatar, Oman and Bahrain pathways

RQ8. Where will Saudi EV charging load appear?

Method	Data	Outputs
Distribute transport electricity (elect_td_trn) to provinces by registrations and population; compare proportional, gravity and convergence methods	vehicle registrations (GASTAT); population	provincial EV load 2030–2060; peak-load implications

RQ9. How much does fleet realism change cumulative transport CO_2?

Method	Data	Outputs
Soft-link GCAM with a stock-turnover model (or LEAP); iterate share-weights until new-vintage shares match modelled sales	survival curves, registrations	fleet, sales, stock lag; change in cumulative CO_2 vs stand-alone GCAM

RQ10. Do consumer groups (early adopters to late majority) change BEV uptake under the same carbon price?

Method	Data	Outputs
Split the 4W car sector into consumer-segment pass-throughs with disutility cost adders (McCollum et al. 2017); compare with MESSAGEix-Transport	McCollum et al. 2017 parameters	BEV uptake curves; the gap between price-only and price-plus-charging policies

The Go 3 project

Task. In teams of two or three, design, run and defend a GCAM transport study that answers one decision-relevant question (one of the ten above or your own). You deliver a proposal (week 1 of the project), a report of about 6,000 words with figures, a reproducible package, and a 15-minute presentation.

Deliverable	Contents	Weight
Proposal (2 pages)	question, decision and metric; scenario design matrix; uncertainty plan with sources; figures you will make (kit `templates/research_proposal.md`)	10 %
Report	introduction and literature (module 3.3); methods with every model change; results as differences from Reference; decomposition; sensitivity or scenario discovery; limits and ethics; conclusion for the decision-maker	50 %
Reproducible package	GCAM version and commit; add-ons, configuration or batch, gcamdata mods; query file; scripts; extracted results in IAMC format; run log	20 %
Presentation	15 minutes for a policy audience; one slide on what the model cannot tell us	20 %

Rubric

Criterion	Excellent (A)	Adequate (C)	Weak (F)
Question and design	one decision, one metric; scenarios isolate the effect; uncertainty planned	clear topic, scenarios mostly controlled	topic only; several changes at once
Model use	version stated; every change documented and verified in output; solver checked	changes documented; some not verified	changes unclear or wrong units
Analysis	differences from Reference; decomposition; sensitivity with interpretation	differences shown; limited explanation	levels only; no comparison
Literature	results placed against GCAM and other-model studies with versions	some comparison	none
Limits and ethics	specific to the question; affect the conclusions	generic list	absent
Reproducibility	a reader can rebuild every figure	most files present	results on request
Communication	figures readable alone; conclusion a decision-maker can use	clear but technical	hard to follow

Self-check

Answer without the handbook, then check below. If you miss a question, reread the module in brackets.

Warm-up

1. Two GCAM-KSA results to remember: 2060 transport CO_2 under net zero, and the 2015 level. (3.3, 3.7)

Core

2. Name the four transport final demands in GCAM. (3.2)
3. Which logit form does GCAM use for mode choice, and which for drivetrain choice? Why does the second matter once carbon prices enter? (3.2)
4. Write the money cost per passenger-km of a technology (Eq. 3.3) and say what happens to it if the load factor doubles. (3.2)
5. A v8.2 tutorial gives a non-fuel cost in 1990$/vkm. What two things must you check before using it in v9.1? (3.5)
6. Why can setting a 2025 share-weight in an add-on file fail to change 2040 results? (3.5)
7. Your add-on changed nothing. Give three likely causes. (3.5)
8. A ZEV standard of 40 % of car pkm: what coefficient goes on the credit input, and what happens if ZEVs are already 50 %? (3.5)
9. What does the LMDI method give that a simple "change in each factor × base-year value" does not? (3.6)
10. Why does stock lag sales, and which GCAM output is closest to sales? (3.7)
11. Give three items that must appear in a GCAM paper according to the community guidelines. (3.8)

Challenge

12. Define coverage and density, and say why you cannot maximise both. (3.6)
13. Why does a 30 % BEV cost change move BEV shares more than a \$200/tCO_2 carbon price in the toy example? (3.2, 3.6)
14. In what sense is a downscaled country pathway "not data"? (3.7)
15. A study sets BEV share-weights to 1 from 2025. Is it a policy scenario? What would make it one? (3.4, 3.5)

Answers

Try it now

- **3.1** (1) In v9.1, `exe/configuration_ref.xml` loads `../input/gcamdata/xml/transportation_UCD_CORE.xml` (name "transportation"); the next line loads `Ccoef.xml` (carbon content). (2) GCAM updates an object only if it already exists when the add-on is read; listed earlier, the add-on would be overwritten by the core file or create objects the core file then replaces.
- **3.2** (1) Money cost rises from 0.286 to 0.334 $/pkm (+17 %), because every per-vehicle cost is shared among fewer people. (2) $e^{-8\Delta/p_0} = 0.8/0.2 = 4 \Rightarrow \Delta/p_0 = -\ln 4 \div 8 = -0.173$: the BEV must be about 17 % cheaper. (3) Fuel is only about 5 % of the petrol car's money cost per pkm (Figure 3.3), so a carbon price changes the total little; capital is about three-quarters of it.
- **3.3** Open-ended. A good pairing for (2) is Edelenbosch et al. (2017) or Yeh et al. (2017) against a study with behavioural detail (McCollum et al. 2017); evidence that would change your mind is observed mode shift after price or infrastructure changes that the model reproduces.
- **3.4** (1) Example: "Is a 2040 hydrogen-truck mandate or a carbon price the cheaper way to cut Gulf road-freight CO_2 by 40 %, and how sensitive is the answer to electrolyser cost?" (2) 2 × 3 × 2 × 3 = 36 runs; drop or reduce the logit-exponent factor first (test it only on the central case), because it is a structural sensitivity, not a policy lever.
- **3.5** (1) `region name="Middle East"` → `supplysector name="trn_pass"` → `tranSubsector name="Passenger Rail"` → `<speed year="2040">60</speed>`. (2) 0.30 × 0.469 = 0.141 1975$/vkm. (3) Every truck technology in the freight road sector demands 0.2 credits per unit of output; BEV and FCEV trucks supply 1; a `policy-portfolio-standard` of type RES clears the market.
- **3.6** (1) Zero if pkm is constant ($\ln 1 = 0$); negative if pkm falls. (2) Density 0.3 means 70 % of runs in the box are fine: the box is too broad. Add a second input or tighten the bounds, and report the coverage you lose. (3) BEV cost — the longest bar — or the share-weight path, because it is an assumption with no data behind it.
- **3.7** (1) With convergence by 2040, w = 1 from 2040, so the 2050 share is the target, 0.42, and stays there. (2) Higher: with a 12-year life the stock averages fewer, more recent (higher-share) years — 0.18 instead of 0.15. (3) GCAM → LEAP: "transport service output by mode" and fuel prices; LEAP → GCAM: an add-on with `<share-weight>` or `<loadFactor>` values per period.
- **3.8** (1) e.g. "We use a modified version of GCAM v9.1 (GCAM-<institution>-TRN), an open-source multisector model developed at JGCRI (JGCRI 2025), with 32 energy-economy regions … Our changes to the Middle East transport sector are listed in Table S1." (2) The configuration or batch file, the carbon-tax add-on, any transport add-ons, the query file, and the analysis script (plus the run log). (3) For a ZEV mandate: "GCAM has no vehicle stock, so the mandate is modelled as a share of passenger-km, and fleet effects lag sales by about a decade."

Self-check

1. 115 Mt in 2060 under net zero (Kamboj et al. 2024 DP); 176 Mt in 2015.
2. Passenger (trn_pass), freight (trn_freight), international aviation, international shipping.
3. Relative-cost logit for modes; absolute-cost logit for drivetrains. The absolute form handles costs near zero or negative, as with subsidies and credit prices.
4. $P = (P_f \cdot I + N) \div L$. Doubling L halves the money cost and energy per pkm (the time cost is unchanged).
5. Units (1990$ → 1975$, and BTU → EJ per billion vkm for intensities) and the object names (e.g. hydrogen inputs renamed in v8.8).
6. The default interpolation rule recomputes later share-weights; add an interpolation-rule (e.g. fixed) in the same file.
7. Listed before the core file; a misspelt or outdated name; an interpolation rule overriding it (or old units).
8. 0.4; at 50 % ZEV the constraint does not bind and the credit price is zero.
9. An exact split with no residual, symmetric in the two years (or scenarios).

10. Vehicles last 12–15+ years, so the stock averages many years of sales. "transport service output by tech (new)".
11. The GCAM version; a description of every modification (and a renamed model for substantial changes); the repository and DOI of the version used. Also: cite JGCRI (2025).
12. Coverage = high in box ÷ all high; density = high in box ÷ all in box. Enlarging the box catches more high runs (coverage up) but also more fine ones (density down).
13. Capital is most of the cost per pkm and fuel a small part; the carbon price only acts on fuel.
14. It combines the regional run's assumptions with the method's (e.g. the convergence year); it is a model result, not an observation.
15. No — it is an assumption about preferences. Pairing it with a policy (credit standard, subsidy) whose cost GCAM reports makes it a policy scenario.

Glossary

Term	Meaning
absolute-cost logit	shares depend on cost differences scaled by a base cost p_0; used for drivetrain choice (Eq. 3.6)
add-on file	an XML file read after the core inputs that changes only what it names
batch mode	GCAM option that runs every combination of component sets listed in a batch file
Broyden's method	the quasi-Newton method GCAM's solver uses to clear all markets in log prices
calibration	solving base-year share-weights so the model reproduces observed energy use
configuration file	the list of input XML files (and settings) for one run, in reading order
coverage / density	share of all high runs inside a scenario-discovery box / share of runs in the box that are high
credit market	a policy-portfolio-standard market in which some technologies demand and others supply credits
DSCALE	open-source method that downscales regional pathways to countries by convergence (Sferra et al. 2026)
final demand	the four transport services: passenger, freight, international aviation, international shipping
gcamdata	the R package that turns source data into GCAM input XML files
GCAM-KSA	GCAM 6.0 with Saudi Arabia as its own region (KAPSARC and PNNL); not in the public release
income / price elasticity	% change in service demand per % change in income / service price (1.0 and −1.25 for trn_pass in core)
interpolation rule	how GCAM fills share-weights (and other values) between specified years
LMDI	logarithmic mean Divisia index: splits a change into factor effects with no residual (Eq. 3.9)
load factor	people (or tonnes) per vehicle; divides all per-vehicle costs and energy
non-fuel cost	per-vehicle-km cost of owning, insuring and maintaining a vehicle (1975$/vkm from v8.5)
OTAQ/EMF37 table	harmonised vehicle costs and efficiencies that override the UC Davis values for most modes
recursive-dynamic	solved period by period with no foresight of later periods
relative-cost logit	shares depend on cost ratios; used for mode choice (Eq. 3.6)
service output by tech (new)	output of the newest vintage: GCAM's nearest proxy for sales
share-weight	the calibrated preference parameter α in the logit; absorbs everything the costs miss
soft link	two models exchanging numbers through files and iterating to agreement
time-value multiplier	fraction of the wage counted as the cost of an hour spent in a mode
UCD database	UC Davis transport database: costs, intensities, load factors and speeds for 14 regions
value of time	wage × multiplier ÷ speed: the time cost per pkm added to each mode's price (Eq. 3.5)

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Honesty note

What was checked. GCAM structure, file names, element names, logit types, exponents, elasticities and units were read from the gcam-core repository at tag gcam-v9.1 and compared with older tags; documentation statements come from the GCAM v8.2 documentation (checked 23 October 2026). Most DOIs were confirmed against publisher or repository records during the research for this go. Every number in the worked examples was computed in Python from the repository data or from stated illustrative inputs.

Illustrative, invented for teaching: fuel prices in the cost build-up (Figure 3.3); bus and rail money costs and mode share-weights (Figure 3.5); the LMDI country; the tornado inputs; the 36-run scenario-discovery example; the downscaling shares; the 15-year rectangular fleet lifetime; the toy-model region and all its results.

Not verified, or flagged in the research: the DOI of the KAPSARC transport discussion paper (KS--2024-DP01) and of Qiu et al. (2024) in *ERL*; some page numbers and the full model lists of a few multi-model papers; the contents of the GCAM-KSA documentation (registration required); the licence of the KAPSARC data portal; whether a transport-only carbon price can be set by renaming the CO_2 object; whether negative `output-ratio` values are allowed for intensity standards; the exact column naming of year columns in gcamdata user-modification chunks; the gcamreader `q.title` attribute. DSCALE is presented through a simplified convergence formula, not its full algorithm.

Not run against a live GCAM installation: all XML templates, configuration and batch files, and the GCAM reader scripts in the kit. Two kit assumptions in particular need a test run: the ZEV-standard file scales credit coefficients with an assumed load factor of 1.6, and the batch file uses empty FileSets for its Reference levels. The toy model and analysis scripts run on synthetic or toy data.

Kit files quoted: `analysis/decompose_lmdi.py`, `analysis/out/scenario_discovery.txt`, `analysis/read_transport.R`, `analysis/read_transport.py`, `analysis/scenario_discovery.py`, `config/batch_transport.xml`, `config/configuration_transport_snippet.xml`, `downscale/downscale_demo.py`, `templates/reporting_checklist.md`, `xml/policy_carbon_tax.xml`, `xml/policy_co2_constraint_netzero.xml`, `xml/policy_zev_standard.xml`, `xml/trn_bev_cost_low.xml`, `xml/trn_freight_truck_efficiency.xml`, `xml/trn_loadfactor_up.xml`, `xml/trn_zev_shareweights.xml`.